

The Most Informative Bit and Beyond: A Proof of the Courtade–Kumar Conjecture and Multibit Extensions

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Abstract—Let $X = (X_1, \dots, X_n)$ be uniform on the Boolean cube, and let Y be obtained by passing its coordinates independently through a binary symmetric channel with crossover probability p . A natural quantization question asks how much information about Y can be retained by a function $Q(X)$ constrained to k output bits. For $k \leq n$, reporting k input coordinates retains $k(1 - h_2(p))$ bits of information, where h_2 denotes the binary entropy function. This makes coordinate projections the natural benchmark.

The case $k = 1$ is the Courtade–Kumar conjecture, which posits that every Boolean function f satisfies

$$I(f(X); Y) \leq 1 - h_2(p),$$

with equality attained by coordinate functions. We prove this conjecture for every dimension and crossover probability, without any balance assumption on f . Our proof views the gap relative to the coordinate benchmark as a quantity that evolves with the channel correlation. After a monotone rearrangement, an entropy-flow identity relates its derivative to the edge boundary of the corresponding Boolean decision set. Combining a sharp local entropy comparison with Fourier analysis on the Boolean cube and logarithmic Sobolev estimates for pivotal sets yields a global differential inequality: if the gap is positive at any correlation level, then it must grow as the noise decreases. This contradicts the noiseless endpoint, where the gap is nonpositive.

For $k \geq 8$, however, a distinctly coding-theoretic behavior emerges. We construct quantizers from shortened Hamming codes whose covering structure beats the coordinate-projection benchmark for $k = 8$, and a direct-sum argument extends this failure to every $k \geq 8$. Thus, unlike in the single-bit case, structured coding can retain more information than simple coordinate selection. Finally, exhaustive searches within several structured code families and Monte Carlo searches over random codebooks for $2 \leq k \leq 7$ reveal a clear advantage of structured codebooks over random ones while still underperforming the coordinate benchmark. These findings lead us to conjecture that coordinate projections remain optimal for $2 \leq k \leq 7$.

Index Terms—Binary symmetric channel, Boolean functions, Courtade–Kumar conjecture, Fourier analysis, Hamming codes, logarithmic Sobolev inequality, mutual information, quantization.

I. INTRODUCTION

Let $X = (X_1, \dots, X_n)$ be a sequence of independent uniform binary random variables, and let Y be obtained by passing every coordinate through a binary symmetric channel (BSC) with crossover probability $p \in [0, 1/2]$. A Boolean

summary $f(X)$ retains one bit from the input. Courtade and Kumar conjectured that the mutual information between this summary and the complete noisy observation is maximized by retaining one of the original coordinates [1]. Equivalently,

$$I(f(X); Y) \leq 1 - h_2(p), \quad (1)$$

where h_2 is the binary entropy function. Equality is attained by any *dictator* defined as $f(X) = X_i$ for some $i \in \{1, 2, \dots, n\}$.

The problem is different from maximizing correlation or noise stability. The observation Y is an n -dimensional random vector, and information carried jointly by its coordinates need not decompose into coordinatewise contributions. For example, the parity of two noisy bits can contain information that is absent from either noisy bit alone. Consequently, results for two Boolean outputs [2], [3] or for sums of coordinatewise mutual informations [4] do not directly imply (1). The output $f(X)$ may also be biased, which removes several symmetries available in the balanced case.

This paper establishes (1) in full generality. The proof uses the correlation parameter $\rho = 1 - 2p$. After monotone sorting, the mutual information becomes an entropy functional of the posterior mean $T_\rho f$. Its derivative admits an edge decomposition. The central step is to strengthen the entropy-constrained edge envelope by retaining the squared angular displacement lost in a Cauchy–Schwarz comparison. The pivotal sets of the same Boolean function supply the Fourier and logarithmic-Sobolev control needed to pay for this correction. The final argument is dynamical: wherever a function has more information than a dictator, its advantage must increase with ρ . Such an advantage cannot return to its nonpositive value at $\rho = 1$.

We also study the natural vector-valued extension in which $f(X)$ contains k bits. The scalar theorem does not tensorize through the chain rule because conditioning on preceding output bits destroys the uniform input law. In fact, the coordinate benchmark is false for sufficiently large fixed k . We give explicit shortened-Hamming counterexamples for $k = 8, 9, 10$, extend them to every $k \geq 8$, and formulate a threshold conjecture for $2 \leq k \leq 7$.

A. Related Work

Courtade and Kumar introduced the full-vector problem and proved several structural reductions, including the monotone sorting operation used below [1]. The conjecture stimulated

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a sequence of Fourier, functional-inequality, and information-geometric approaches. The Friedgut–Kalai–Naor theorem identifies the rigidity of Boolean functions whose Fourier mass is concentrated on the first two levels [5]. Samorodnitsky established the conjecture in a dimension-free high-noise regime and developed strengthened inequalities for the entropy of noisy Boolean functions [6], [7]. Ordentlich, Shayevitz, and Weinstein obtained sharper balanced high-noise bounds using Fourier analysis and hypercontractivity [8], while Yang and Wesel gave a complementary calculus argument in a dimension-dependent neighborhood of complete noise [9].

Yu developed a general Φ -stability framework and subsequently proved broad local-optimality and balanced finite-range results using quantitative Fourier estimates [10]–[12]. Javanmard and Woodruff proved the dictator bound for the sum of the individual-coordinate mutual informations and sharpened the high-noise analysis [4]; the related Boolean-hull and moment relaxations are discussed in [13]. Li and Médard connected the problem to posterior moments and noninteractive correlation distillation [14], and Barnes and Özgür showed that the balanced conjecture is essentially equivalent to a symmetrized Li–Médard formulation [15].

A second line of work seeks stronger entropy or isoperimetric statements. Anantharam, Bogdanov, Chakrabarti, Jayram, and Nair proposed a Hellinger inequality implying the conjecture [16]. Chen and Nair placed this proposal in an ordered family of entropy inequalities and related the low-noise limits to sharp cube isoperimetry [17]. The critical square-root isoperimetric inequality was recently established by Durcik, Ivanišvili, Roos, and Xie [18]. These limiting or stronger statements do not themselves yield the finite-noise Shannon inequality.

The closest antecedent to the present proof is the differential-equation framework of Chen, Gohari, and Nair [19]. Their entropy-flow identity and entropy-constrained two-point extremal formula are used here.

Two concurrent works establish the scalar conjecture in full generality [20], [21]. Chen, Gohari, Javanmard, Lin, Mirrokni, Nair, and Woodruff close the differential-equation program through an unrestricted four-moment Bellman inequality for a hybrid potential, using analytic reductions together with certified interval arithmetic. Ky and Tran instead develop an entropy-production and spectral framework. Their method is substantially closer to ours: both arguments use monotone compression, edgewise entropy-production estimates, Fourier control of functions with diffuse coordinate dependence, separate channel regimes, and a differential contradiction at the noiseless endpoint. The quantitative implementations differ. Our proof centers on an angular edge correction and logarithmic-Sobolev estimates for pivotal sets, whereas Ky and Tran use a profile-clock integration and selected-coordinate spectral estimates. Finally, neither of [20], [21] considers the k -bit extension studied in the present paper where we conjecture that up to $k = 7$, the most informative k -bit function follows a similar structure.

Related operator and channel perspectives include principal inertia components [22] and mutual-information bounds based on adjacency events [23]. Continuous analogues and stochastic-

interpolation variants were developed in [24], [25].

The present paper additionally develops the vector-valued extension in which the quantizer reports k bits. For this problem, Chandar and Tchamkerten characterized the positive-rate asymptotics and gave Hamming- and Golay-code counterexamples with $k \geq 11$ [26]. Huleihel and Ordentlich proved coordinate optimality at the complementary endpoint $k = n - 1$ [27]. The finite-block behavior observed in our code search is also related qualitatively to recent results showing that structured Reed–Solomon-based codebooks can outperform random codebooks under average Hamming covering radius [28]; the objective in that work is different from BSC mutual information.

B. Contributions and Organization

Our first main contribution is Theorem 2, which proves the Courtade–Kumar conjecture in every dimension and for every crossover probability, without any balance assumption on the Boolean function. The proof follows a single dynamical strategy. For $I_f(\rho) := I(f(X); Y_\rho)$, $\phi(\rho) := I(X_1; Y_\rho)$, $\Delta_f(\rho) := I_f(\rho) - \phi(\rho)$, we show that a positive advantage over a coordinate projection would have to grow strictly as the channel becomes less noisy. Section II introduces the posterior representation of I_f , the required Fourier notation, and the monotone reduction. In particular, Lemma 3 shows that coordinatewise sorting preserves any advantage, allowing the remainder of the proof to exploit the edge structure of a monotone Boolean decision set. An entropy-flow identity expresses $\rho I'_f(\rho)$ as an action functional of the posterior mean $T_\rho f$. Section III then decomposes this action over edges of the discrete cube. The key estimate, Lemma 5, strengthens the usual two-point entropy envelope by retaining a squared angular correction that is exact for dictators. The rest of the proof shows that the Boolean and spectral structure of f always provides enough reserve to pay for this correction.

Writing $\ell = \operatorname{arctanh} \rho$, we divide the channel into the small-correlation range $0 \leq \ell \leq 31/20$, the intermediate range $31/20 \leq \ell \leq 7/2$, and the near-noiseless tail $\ell \geq 7/2$. Equivalently, the two junctions are $\rho = \tanh(31/20)$ and $\rho = \tanh(7/2)$. These cutoffs are technical junctions at which the corresponding estimates meet, rather than asserted phase transitions in the problem. Proposition 11 settles the small-correlation range directly, at every output bias, by combining a Fourier energy gap with scalar entropy bounds. In the intermediate range, the pivotal sets of f supply the needed reserve: logarithmic Sobolev inequalities control the noisy entropies of their indicators, and a log-sum argument combines the coordinatewise costs with the global edge budget. This yields Proposition 6, according to which $\Delta_f(\rho) > 0$ implies $\Delta'_f(\rho) > 0$ throughout that range. The bounded angular reserve is insufficient by itself near the noiseless endpoint, so the tail argument supplements the same edgewise inequality with degree-weighted Fourier information. Lemma 7 identifies the relevant coordinate deficits, Lemma 9 converts the information advantage into an entropy-funded spectral baseline, and Lemma 10 turns the resulting budget comparison into strict information growth. Proposition 12 separately

rules out a counterexample having a sufficiently influential coordinate. Section V assembles these complementary estimates in Lemma 13, which gives the global implication

$$\Delta_f(\rho) > 0 \implies \Delta'_f(\rho) > 0.$$

Since $\Delta_f(1) = H(f(X)) - \log 2 \leq 0$, a positive advantage cannot persist to the noiseless endpoint; this contradiction completes the proof of Theorem 2.

Our second contribution, developed in Section VI, concerns the natural extension in which the quantizer reports k bits and the coordinate benchmark is $k(1 - h_2(p))$. Proposition 14 shows that affine linear maps cannot violate this benchmark. Nonlinear nearest-codeword quantizers built from linear codebooks behave differently: an exact high-noise expansion, expressed through the coset-leader distribution, produces shortened-Hamming counterexamples for $k = 8, 9, 10$ in Proposition 15. A direct-sum construction in Proposition 16 extends the failure to every $k \geq 8$. For $k < 8$, exhaustive searches over several structured code families and Monte Carlo experiments with random codebooks find no violation, while also showing a clear advantage of structured codebooks over random ones in this finite-block regime. These findings motivate Conjecture 18, which predicts that coordinate projections remain optimal for $2 \leq k \leq 7$. The appendices provide the scalar inequalities, logarithmic-Sobolev and Fourier estimates, channel-range comparisons, endpoint reductions, exact arithmetic certificates, and reproducible numerical diagnostics used throughout the proof.

II. PROBLEM FORMULATION AND PRELIMINARIES

A. Posterior-Mean Representation

Let n be a positive integer, X a uniform sign string in $\{-1, 1\}^n$, and f a function from these strings to $\{-1, 1\}$. Obtain Y by flipping each bit with probability $(1 - \rho)/2$. Correlation $\rho = 0$ means complete noise; $\rho = 1$ means no noise. Define

$$L = \log 2, \quad h(r) = L - \phi(r),$$

$$\phi(r) = \frac{1+r}{2} \log(1+r) + \frac{1-r}{2} \log(1-r).$$

Thus $h(r)$ is the entropy of a sign-valued bit whose mean is r . In particular, $h(0) = L$, $h(\pm 1) = 0$, and $\phi'(r) = g(r) := \operatorname{atanh} r$.

For $m = \mathbb{E}f$, write

$$u(y) = T_\rho f(y) = \mathbb{E}[f(X) \mid Y = y].$$

Constant rules retain no information, so only nonconstant f need be studied. For these, $-1 < u(y) < 1$ when $0 < \rho < 1$, since every input string has positive conditional probability. The conditional probabilities of $f(X)$ are $(1 + u(y))/2$ and $(1 - u(y))/2$. The mutual information is therefore

$$I_f(\rho) = h(m) - \mathbb{E}h(u) = \mathbb{E}\phi(u) - \phi(m). \quad (2)$$

Every expectation without a subscript is uniform on the appropriate cube. All logarithms are natural, so information is measured in nats. This is the full-vector objective of Courtade and Kumar [1]. It differs from the two-Boolean-output objective

settled in [3] and the sum of coordinate informations bounded in [4], [13]: the conditional expectation above retains the entire observation Y .

Remark 1 (Dictator normalization). If $f(x) = x_1$, then $m = 0$, $u(y) = \rho y_1$, and

$$I_f(\rho) = L - h(\rho) = \phi(\rho).$$

The target is $I_f(\rho) \leq \phi(\rho)$. The parameter ρ is the correlation, not the crossover probability.

Theorem 2 (Courtade–Kumar inequality). *For every positive integer n , every Boolean function $f : \{-1, 1\}^n \rightarrow \{-1, 1\}$, and every $0 \leq \rho \leq 1$,*

$$I(f(X); Y) \leq \phi(\rho). \quad (3)$$

B. Boolean Fourier Representation and Entropy Flow

Write $[n] = \{1, \dots, n\}$. For a subset $S \subseteq [n]$, its size is $|S|$; the empty subset is $\emptyset$. Define $\chi_S(x) = \prod_{i \in S} x_i$, with empty product one. The Fourier coefficient of any cube function v is $\hat{v}(S) = \mathbb{E}[v(X)\chi_S(X)]$. The parity functions form an orthonormal basis under the uniform measure. Thus

$$f = \sum_{S \subseteq [n]} \hat{f}(S) \chi_S, \quad \mathbb{E}f^2 = \sum_S \hat{f}(S)^2.$$

A degree- $|S|$ parity survives independent noise with factor $\rho^{|S|}$, so

$$T_\rho f = \sum_S \rho^{|S|} \hat{f}(S) \chi_S.$$

This is the cube instance of the conditional-expectation spectral viewpoint developed through principal inertia components in [22]. Here we know each eigenvalue explicitly and keep its Fourier degree; the standard Boolean Fourier facts used below are reviewed in [29]. Define the cube Laplacian $\mathcal{L}\chi_S = |S|\chi_S$. Write $x^{\oplus i}$ for x with coordinate i flipped. In physical coordinates,

$$\mathcal{L}v(x) = \frac{1}{2} \sum_i (v(x) - v(x^{\oplus i})).$$

Thus $\mathcal{L}$ measures how a field differs from its neighbors, and also multiplies each Fourier coefficient by its degree. With y fixed, $\partial_\rho u(y)$ denotes differentiating the posterior $u(y) = T_\rho f(y)$ with respect to ρ . This partial-derivative notation holds the other inputs fixed. Differentiating the Fourier expansion shows that $\rho \partial_\rho u = \mathcal{L}u$. Likewise $I'_f(\rho) = \frac{d}{d\rho} I_f(\rho)$ is the rate of change of the scalar mutual information with f fixed. Define the action, or rescaled information growth rate, by

$$A(u) := \mathbb{E}[g(u)\mathcal{L}u] = \rho I'_f(\rho). \quad (4)$$

For a dictator, $A = \rho g(\rho)$. The corresponding entropy-flow identity is also central to Chen–Gohari–Nair [19, Lemma 1]; we give the short derivation here to fix normalization.

The following monotone reduction removes changes of direction from the subsequent edge analysis. It acts simultaneously at every noise level.

Lemma 3 (Sorting preserves a possible advantage). *Every Boolean function can be replaced by a monotone Boolean*

function in the same dimension, with the same mean and at least as much information at every correlation.

Proof. Sort the function without decreasing its information. This reduction is due to Courtade and Kumar [1]; here is the argument in our notation. In a chosen coordinate, the sections are the two functions obtained by fixing that input bit to $+1$ or -1 . Let a be their average and b their half-difference, so the sections are $a + b, a - b$. Replace them by $a + |b|, a - |b|$. The mean is preserved. In this paragraph T denotes noise averaging on the other coordinates at correlation ρ . After applying T , the symmetric pair has half-separation $\rho T b$, replaced by $\rho T |b|$. Positivity gives $|T b| \leq T |b|$. For any convex function, its average at a symmetric pair increases with the absolute separation. Thus sorting cannot decrease I_f , at any noise level. Min and max preserve previously sorted coordinates, so sorting all coordinates leaves a monotone function. Each sorting operation works simultaneously at all correlations, so the same sorted function has the claimed advantage throughout $[0, 1]$. $\square$

Here *monotone* means that changing an input coordinate from -1 to 1 never decreases the output. The sorting step is what allows Boolean structure to enter the next two lemmas. Write $\Delta_f(\rho) = I_f(\rho) - \phi(\rho)$ for the lead over the dictator. It is continuous on $[0, 1]$ and differentiable inside. At the noiseless endpoint,

$$\Delta_f(1) = h(m) - \log 2 = -\phi(m) \leq 0.$$

The proof will show that a positive lead must be increasing. Such a lead cannot return to zero, contradicting this endpoint bound. Unlike an optimizer analysis such as the clustering formulation in [30], this implication requires no stationarity of f . Unlike a proof confined to a high-noise neighborhood [6], [9], its conclusion is used along a whole interval ending at the noiseless endpoint. The inequalities below must supply that finite-range control.

Use one common boundary for the two arguments:

$$\rho_* := \tanh(31/20).$$

For $0 \leq \rho \leq \rho_*$, Appendix D proves the information bound at every bias, using the elementary Fourier estimate in Section D-A. The growth argument below starts at exactly the same boundary. Thus a positive lead must occur above ρ_* ; no overlap between separately chosen ranges is needed.

For a putative violation, the information premise is exactly

$$\mathbb{E}h(T_\rho f) < h(\rho) - \phi(m). \quad (5)$$

It supplies a positive credit in the spectral baseline. Our target is $A(u) > \rho \ell_\rho$, with $\ell_\rho = \text{atanh } \rho$: by (4), this says that the lead is increasing.

III. EDGE GEOMETRY AND THE PIVOTAL DECOMPOSITION

For a coordinate index i , let $f_{i,+}$ and $f_{i,-}$ be the sections obtained by fixing that input to $+1$ and -1 , respectively. For a monotone Boolean function, the derivative in coordinate i , viewed on the other $n - 1$ coordinates, is

$$q_i^f = \frac{f_{i,+} - f_{i,-}}{2} \in \{0, 1\}.$$

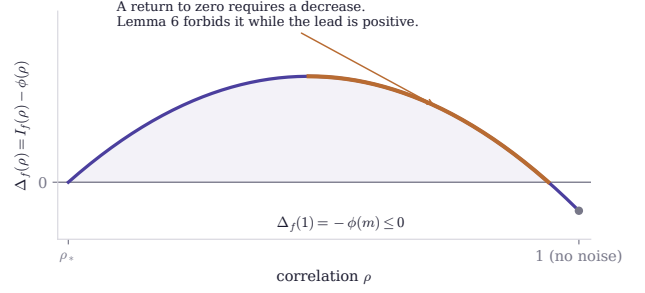


Fig. 1. Schematic of the growth contradiction. A positive lead would have to decrease before reaching its nonpositive value at $\rho = 1$, whereas Lemma 13 shows that it is strictly increasing whenever it is positive.

It indicates a pivotal edge: one on which the output changes. The influence $\text{Inf}_i(f)$ is the probability of such an edge. Its mean

$$a_i = \mathbb{E}q_i^f = \widehat{f}(\{i\}) = \text{Inf}_i(f)$$

is both the singleton Fourier coefficient and the influence. This identity is special to monotone Boolean functions. After noise, $\mathbb{E}(u_{i,+} - u_{i,-})/2 = \rho a_i$. For a dictator, $f_{1,+} = 1$ and $f_{1,-} = -1$, so $q_1^f = 1$ everywhere; every other q_i^f is zero. Retaining the indicator q_i^f , rather than just its mean, will be essential. Moment relaxations of the posterior can admit objects that are not Boolean noise images, a distinction also examined in [13, Section 8.1.2]. Our logarithmic Sobolev step acts on the actual pivotal indicator and its positive smoothings.

For an edge with endpoint values p, q , the action and angle energy contributions are

$$a(p, q) = \frac{(p - q)(g(p) - g(q))}{4},$$

$$j(p, q) = \frac{(\arcsin p - \arcsin q)^2}{4}.$$

Average over the edges in each coordinate direction, then sum over directions. The results are $A(u)$ and $\mathbb{E}[\Theta \mathcal{L} \Theta]$, with $\Theta = \arcsin u$. Cauchy–Schwarz gives $a \geq j$. We need a stronger statement which keeps track of the difference.

Remark 4 (Angular parametrization). The derivative of g is $g'(r) = 1/(1 - r^2)$. Its square root is the derivative of $\arcsin r$. For $p \geq q$,

$$\left(\int_q^p \frac{dr}{\sqrt{1 - r^2}} \right)^2 \leq (p - q) \int_q^p \frac{dr}{1 - r^2}.$$

This is exactly $j(p, q) \leq a(p, q)$. The angle turns an edge's information cost into a squared difference, which Fourier analysis can handle. The lemma retains the cost left over in this conversion.

Define

$$F(v) = \frac{h(\tanh v)}{\tanh v}, \quad c(v) = \frac{\arcsin(\tanh v)^2}{\tanh v}.$$

The function F decreases from infinity to zero. An entropy budget $H_{\text{budget}} > 0$ sets the centered comparison pair for each active coordinate: its values are $\pm \tanh v_i$, where

$$F(v_i) = \frac{H_{\text{budget}}}{\rho a_i}.$$

This matches entropy per unit width. The comparison height v_i is not a posterior observed on an individual edge. A smaller entropy budget gives a larger height and a stronger growth bound.

The first bound below is the entropy-constrained edge envelope of Chen–Gohari–Nair [19, Theorem 4 and Appendix C], written for signs and natural logarithms. We also need the angle correction, whose derivation is supplied in Appendix A. No conjectured inequality from that paper is assumed. The difference from its potential-induction program is where the remaining cost is paid: here the edge correction is combined with Fourier information from the same Boolean rule.

The same centered comparison pair will control both the total growth and the residual growth after subtracting the angle energy. Using a common entropy budget in the two bounds is essential for combining them later.

Lemma 5 (The edge budget). *Let f be nonconstant and monotone, $0 < \rho < 1$, and $u = T_\rho f$. Choose any entropy budget $H_{\text{budget}} \geq \mathbb{E}h(u) > 0$. For every coordinate with $a_i > 0$, define v_i by $F(v_i) = H_{\text{budget}}/(\rho a_i)$. Then*

$$A(u) \geq \rho \sum_i a_i v_i, \quad (6)$$

$$A(u) \geq \mathbb{E}[\Theta \mathcal{L} \Theta] + \rho \sum_i a_i (v_i - c(v_i)). \quad (7)$$

Coordinates with $a_i = 0$ contribute zero and are omitted.

Proof. For one edge put $d_{\text{edge}} = |p - q|/2$ and $H_e = [h(p) + h(q)]/2$. The notation F^{-1} means the inverse of the decreasing profile F , not its reciprocal. Define the correction function corr by $\text{corr}(F(v)) = v - c(v)$. The scalar calculation in Appendix A proves

$$\begin{aligned} a(p, q) &\geq d_{\text{edge}} F^{-1}(H_e/d_{\text{edge}}), \\ a(p, q) - j(p, q) &\geq d_{\text{edge}} \text{corr}(H_e/d_{\text{edge}}). \end{aligned}$$

Both right sides are convex in (d_{edge}, H_e) and decrease with H_e . For coordinate i , $\mathbb{E}d_{\text{edge}} = \rho a_i$ and $\mathbb{E}H_e = \mathbb{E}h(u) \leq H_{\text{budget}}$. Jensen therefore bounds the two averaged edge costs below by their values at $(\rho a_i, H_{\text{budget}})$. Their inverse parameter is exactly v_i . Summing over coordinates proves both claims. $\square$

For a dictator, choose its actual entropy $H_{\text{budget}} = h(\rho)$. Put $\theta = \arcsin \rho$. The sole height is $v_1 = \text{atanh } \rho$, and the two conclusions become $A = \rho \text{atanh } \rho$ and $A = \theta^2 + (\rho \text{atanh } \rho - \theta^2)$. Thus the lemma preserves the complete edge account, including the positive correction.

We use this lemma with two budgets. Whenever $\mathbb{E}h(u) \leq h(\rho)$, the choice $H_{\text{budget}} = h(\rho)$ defines heights ℓ_i by

$$F(\ell_i) = F(\ell_\rho)/a_i. \quad (8)$$

Since $a_i \leq 1$, these satisfy $0 < \ell_i \leq \ell_\rho$. The middle argument will use a smaller budget that also retains the output mean; the lemma applies without changing its proof.

For the following illustration only, v is an edge-center parameter and $k > 0$ its half-width in inverse-hyperbolic-tangent coordinates: $p = \tanh(v + k)$, $q = \tanh(v - k)$. These

are local plot parameters. The plotted d_{edge}, H_e, j are the edge half-separation, average entropy, and angle cost defined above.

With this budget, the first inequality bounds the total comparison weight:

$$W := \sum_i a_i \ell_i \leq A(u)/\rho. \quad (9)$$

For a dictator there is one active coordinate, $a_1 = 1$, $\ell_1 = \ell_\rho$, and equality holds. The second inequality will be combined with a spectral lower bound. Keeping its correction is essential.

IV. INFORMATION GROWTH ACROSS CHANNEL REGIMES

A. Intermediate-Correlation Growth

We first prove the required growth inequality on the intermediate range $31/20 \leq \ell = \text{atanh } \rho \leq 7/2$. In this section write $r = \rho$. The idea is to keep the coordinates together. Logarithmic Sobolev gives an entropy cost for their pivotal probabilities; the log-sum inequality then combines that cost with the edge budget. We will not choose a comparison Fourier degree or divide coordinates by influence. The underlying entropy tool is Gross's cube logarithmic Sobolev inequality [29], [31]. Its use here is specific: the positive kernel in the next calculation converts pivotal-set entropy into the same spectral expression produced by the centered angle comparison. This supplies the correction left after the entropy-flow edge bound [19], without a conjectured coupling potential.

Proposition 6 (Pivotal entropy forces faster growth). *Let f be monotone Boolean and $31/20 \leq \text{atanh } \rho \leq 7/2$. Then*

$$I_f(\rho) > \phi(\rho) \implies I'_f(\rho) > \phi'(\rho).$$

The mean of f is unrestricted.

The proof compares the energy demanded by the information premise with the growth available along the cube edges. It first obtains an angle-energy requirement, expresses part of that requirement as pivotal entropy, and then combines the entropy with the edge correction through the log-sum inequality.

Put $\Theta = \arcsin u$ and $J_{\text{ang}} = \mathbb{E}[\Theta \mathcal{L} \Theta]$. The dictator has posterior values $\pm r$. Define its angle and the slope of the line through those two angle values by

$$\theta = \arcsin r, \quad \lambda = \theta/r.$$

Two further coefficients come from the tangent inequality below:

$$\alpha = \frac{\theta + \sinh \ell}{\ell}, \quad \gamma = \lambda \alpha, \quad \eta = \frac{\theta^2 \gamma}{\gamma + 1}.$$

These are channel functions, not parameters to optimize. The same γ will serve throughout the growth proof, including the high-correlation extension. Only the way we estimate the pivotal cost changes with the channel.

1) *Angle energy required by an information lead:* The following scalar inequality is proved in Appendix B:

$$\begin{aligned} \frac{\alpha}{2} (\arcsin x - \lambda x)^2 &\leq x \arcsin x - r \theta \\ &\quad - \alpha [\phi(x) - \phi(r)], \quad \text{quad} - 1 \leq x \leq 1. \end{aligned} \quad (10)$$

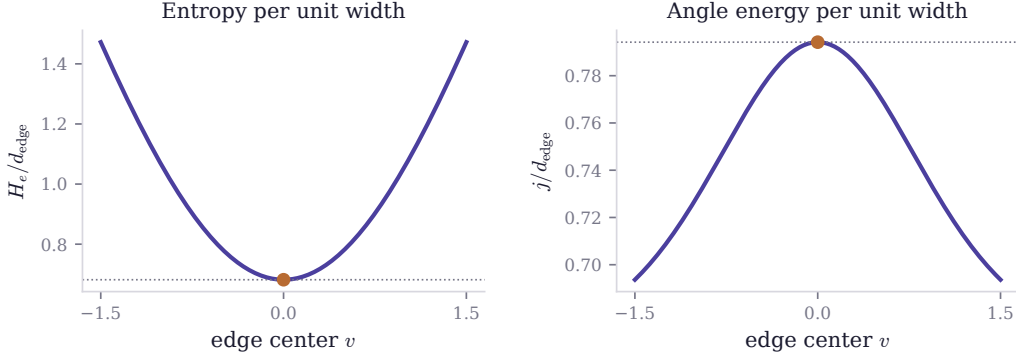


Fig. 2. The geometry behind the edge lemma. Write the endpoints as $p = \tanh(v + k)$, $q = \tanh(v - k)$, with $k = 0.8$ in this illustration. Centering at $v = 0$ minimizes entropy per unit width and maximizes angle energy per unit width. These two extremal facts give a convex lower bound on the action left after angle energy is removed.

Both sides vanish at $x = \pm r$. It bounds the error of the dictator's angle line by an entropy difference, without losing equality at the reference posterior. The comparison stays within Shannon entropy. Although the angle and the Hellinger program [16], [17] both involve the geometry of binary probabilities, we do not infer this inequality from a Hellinger-optimality conjecture; its scalar proof is given in full.

Average (10) with $x = u$, multiply by 2λ , and complete the square in the positive operator $\mathcal{L} + \gamma$. This gives

$$J_{\text{ang}} \geq 2\theta^2 + 2\gamma[\phi(m) + \Delta_f(r)] + \lambda^2 \mathbb{E} \left[u \left\{ \gamma - \frac{(\gamma+1)^2}{\mathcal{L} + \gamma} \right\} u \right]. \quad (11)$$

Here division by $\mathcal{L} + \gamma$ means multiplication of each degree- k Fourier coefficient by $1/(k + \gamma)$. For completeness, the discarded nonnegative square is

$$\mathbb{E}[(\Theta - \lambda(\gamma+1)(\mathcal{L} + \gamma)^{-1}u)(\mathcal{L} + \gamma) \cdot (\Theta - \lambda(\gamma+1)(\mathcal{L} + \gamma)^{-1}u)].$$

The constant Fourier mode is included; this matters when $m \neq 0$. For the dictator, $u = ry_1$ and $\Theta = \lambda u$ are both degree one. The operator $(\gamma+1)(\mathcal{L} + \gamma)^{-1}$ therefore fixes u , so the discarded square is zero. In (11) the spectral term is $-\theta^2$; the result is exactly $J_{\text{ang}} = \theta^2$.

2) *Contribution of noisy pivotal sets:* Recall the pivotal indicator q_i^f , with mean a_i . Omit coordinates with $a_i = 0$; every logarithm below then has a positive argument. Let $t \in [0, r^2]$ denote a squared noise correlation and put

$$w_i(t) = T_{\sqrt{t}} q_i^f, \quad s_i(t) = \mathbb{E} w_i(t)^2.$$

Noise here acts on the other $n - 1$ coordinates. The value $w_i(t)$ is the conditional probability that coordinate i is pivotal. Its mean remains a_i ; its second moment $s_i(t)$ records how unevenly that probability is distributed. Complete averaging gives $s_i(0) = a_i^2$, whereas the unsmoothed indicator has second moment a_i . Define an average of these second moments:

$$z_i = \frac{\gamma+1}{\gamma r^2} \int_0^{r^2} [1 - (t/r^2)^\gamma] s_i(t) dt, \quad Z = \sum_i z_i.$$

The weight in this integral is nonnegative and has total mass one. Thus $a_i^2 \leq z_i \leq a_i$: noise preserves the mean and cannot increase the second moment of an indicator.

Write $V_0 = 1 - m^2$ for the Boolean variance and $U = \text{Var}(u)/r^2$ for the normalized noisy variance. Fourier expansion of the integral gives

$$Z = (\gamma+1) \sum_{k \geq 1} \frac{r^{2(k-1)}}{\gamma+k} \sum_{|S|=k} \hat{f}(S)^2, \quad 0 < Z \leq U \leq V_0.$$

The last inequality is Boolean Parseval followed by noise contraction. For a dictator, $w_1(t) = s_1(t) = z_1 = 1$, all other coordinate terms vanish, and $Z = U = V_0 = 1$. Thus $V_0 - Z$ measures a variance reserve that is absent in the equality example. The exact integral identity

$$\frac{\gamma+1}{\gamma} \int_0^{r^2} [1 - (t/r^2)^\gamma] \sum_i \mathbb{E}[w_i(t) \mathcal{L} w_i(t)] dt = (\gamma+1)r^2(U - Z)$$

is checked one Fourier degree at a time: a degree- k coefficient appears in k pivotal derivatives, each of degree $k - 1$.

Cube logarithmic Sobolev, proved in Appendix A, gives

$$\mathbb{E}[w_i \mathcal{L} w_i] \geq \frac{1}{2} s_i \log \frac{s_i}{a_i^2}.$$

For the auxiliary entropy bound, average under the probability density w_i^2/s_i . Jensen applied to $\log(1/w_i)$ gives $\mathbb{E}_* \log w_i \geq \log(s_i/a_i)$, hence $\text{Ent}(w_i^2) \geq s_i \log(s_i/a_i^2)$. Here $\text{Ent}(v) = \mathbb{E}(v \log v) - (\mathbb{E} v) \log(\mathbb{E} v)$; zero-weight points are omitted. Convexity of $s \log(s/a_i^2)$ and the averaging integral therefore give a bound on the *pivotal entropy cost*, defined by

$$E_{\text{piv}} := \sum_i z_i \log \frac{z_i}{a_i^2}.$$

Each summand is nonnegative since $z_i \geq a_i^2$; a noisy pivotal probability that is constant has zero cost. In this notation,

$$(\gamma+1)(U - Z) \geq \frac{1}{2} E_{\text{piv}}. \quad (12)$$

Rewrite (11) in terms of U, Z, V_0 , including its constant mode, and then apply (12). To keep the bias contribution visible, define

$$M(m) := 2\gamma\phi(m) - [\lambda^2(2 - r^2) + \lambda/\alpha]m^2.$$

The required angle energy is now

$$\begin{aligned} J_{\text{ang}} &\geq \theta^2[1 + V_0 + \gamma U - (\gamma + 1)Z] + M(m) + 2\gamma\Delta_f(r) \\ &\geq \theta^2(1 + V_0 - Z) + \frac{\eta}{2}E_{\text{piv}} + M(m) + 2\gamma\Delta_f(r). \end{aligned} \quad (13)$$

Read the last line term by term: the dictator's energy θ^2 , the variance reserve $\theta^2(V_0 - Z)$, the pivotal entropy cost, the bias correction, and the credit $2\gamma\Delta_f(r)$ for extra information. The last credit is strictly positive under a lead. For the dictator, $E_{\text{piv}} = 1 \log 1 = 0$, and every term except θ^2 vanishes.

3) *Log-sum closure of the edge budget:* Suppose the information lead $\Delta = \Delta_f(r)$ is positive, but its growth is no faster than the dictator's: $A(u) \leq r\ell$. We derive a contradiction. Monotonicity gives $a_i \leq 1 - |m| \leq V_0$. Also

$$\mathbb{E}h(u) = h(r) - \phi(m) - \Delta \leq V_0 h(r),$$

since $\phi(m) \geq m^2/2$ and $h(r) < 1/2$. For the latter, $r > 9/10$ and $L < 7/10$ give $h(r) \leq L - r^2/2 < 1/2$. Apply Lemma 5 with $H_{\text{budget}} = V_0 h(r)$. Its heights satisfy

$$F(v_i) = \frac{V_0 h(r)}{ra_i} = \frac{V_0 F(\ell)}{a_i}, \quad 0 < v_i \leq \ell,$$

where the upper bound follows from $a_i \leq V_0$ and the decrease of F . With $H(v) = v - c(v)$, the lemma gives

$$\sum_i a_i v_i \leq A(u)/r \leq \ell, \quad A(u) \geq J_{\text{ang}} + r \sum_i a_i H(v_i). \quad (14)$$

These heights are comparison parameters; they are not additional properties assumed of the field.

Set $\beta = 2r/\eta$. The channel estimates proved in Lemma 22 are

$$\log \frac{vF(v)}{\ell F(\ell)} \geq \beta[H(\ell) - H(v)] \quad (0 < v \leq \ell), \quad (15)$$

$$\theta^2 - rH(\ell) - \eta/2 \geq 0, \quad (16)$$

$$M(m) - rH(\ell)m^2 \geq 0. \quad (17)$$

Their role is simple: the first converts the edge budget to a normalization for log-sum; the other two leave nonnegative reserves.

For that normalization, put $b_i = a_i^2 e^{-\beta H(v_i)}$ and $B_{\text{piv}} = \sum_i b_i$. This choice puts the edge correction inside the entropy logarithm: $\log(z_i/b_i) = \log(z_i/a_i^2) + \beta H(v_i)$. The weights need not sum to one; log-sum keeps track of their total. Equation (15) gives

$$\frac{b_i}{a_i v_i} \leq \frac{V_0}{\ell} e^{-\beta H(\ell)}, \quad B_{\text{piv}} \leq V_0 e^{-\beta H(\ell)}.$$

The second inequality uses the first bound in (14): a rule growing no faster than the dictator has only ℓ units of edge

weight to allocate. Since $z_i \leq a_i$, $H \geq 0$, and $\eta\beta/2 = r$, log-sum now yields

$$\begin{aligned} \frac{\eta}{2}E_{\text{piv}} + r \sum_i a_i H(v_i) &\geq \frac{\eta}{2} \sum_i z_i \log \frac{z_i}{b_i} \\ &\geq \frac{\eta}{2} Z \log \frac{Z}{B_{\text{piv}}} \\ &\geq \frac{\eta}{2} Z \log \frac{Z}{V_0} + rZH(\ell). \end{aligned}$$

The first line joins the two costs by the logarithmic identity just given. The second is log-sum. The third uses the bound on total weight. This is the connection between the information requirement and the available growth: both involve the same noisy pivotal sets.

Finally insert this bound into (13) and (14). Subtract $r\ell = \theta^2 + rH(\ell)$, and use $Z \log(Z/V_0) \geq Z - V_0$, the tangent inequality for $t \log t$:

$$\begin{aligned} A(u) - r\ell &\geq \underbrace{[\theta^2 - rH(\ell) - \eta/2](V_0 - Z)}_{\text{nonnegative variance reserve}} \\ &\quad + \underbrace{[M(m) - rH(\ell)m^2]}_{\text{nonnegative bias reserve}} + \underbrace{2\gamma\Delta}_{\text{positive lead credit}} > 0. \end{aligned} \quad (18)$$

This contradicts $A(u) \leq r\ell$. We have proved

$$\Delta_f(r) > 0 \implies A(u) > r\ell \quad (31/20 \leq \ell \leq 7/2).$$

At a dictator, $a_1 = z_1 = V_0 = 1$ and $v_1 = \ell$. Log-sum has one term and is exact. The variance reserve, bias reserve, and lead credit are all zero. What remains is the exact account

$$\underbrace{r\ell}_{\text{growth}} = \underbrace{\theta^2}_{\text{angle energy}} + \underbrace{rH(\ell)}_{\text{edge correction}}.$$

An information lead would add the strictly positive last term in (18); the other two terms cannot cancel it.

B. High-Correlation Spectral Regime

The intermediate-correlation argument pays for the edge correction from an angle reserve. The dictator's squared angle θ^2 stays below $\pi^2/4$, while its edge correction $\ell - c(\ell)$ grows without bound. That reserve cannot cover the whole tail. For $\ell \geq 7/2$, we keep the same angle inequality, completed square, and penalty, but estimate the spectral term by comparison with a reference Fourier degree.

C. Coordinate Deficit Bounds

Keep the canonical penalty $\gamma = \lambda\alpha$ from the centered angle comparison. We introduce only a comparison degree $d \geq 2$, which may be real rather than integer. In the tail we choose $d = \ell/2 + 5/4$; every coordinate uses the same degree. Define

$$\begin{aligned} \omega(s) &= \frac{\gamma s}{\gamma + s}, \\ K_{\text{LS}} &= \frac{\gamma^2 e^{2d-3}}{2(\gamma + d)(\gamma + 1)}, \\ d_1 &= [\omega(d) - \omega(1)]\rho^2, \\ d_2 &= [\omega(d) - \omega(2)]\rho^4. \end{aligned}$$

The bounded spectral weight $\omega(s)$ will arise from completing a square. We replace $\omega(|S|)$ by the common value $\omega(d)$. One estimate for the resulting loss uses the entropy of the noisy pivotal set, while the other uses the total energy of its indicator. Neither bound dominates uniformly, so both are retained.

Lemma 7 (Two bounds for the coordinate deficit). *Set*

$$D_i = \sum_{S \ni i} \frac{\omega(d) - \omega(|S|)}{|S|} \hat{u}(S)^2,$$

$$D(a) = \min \left\{ K_{\text{LS}} \rho^2 a^2, \frac{d_2}{2} a + \left(d_1 - \frac{d_2}{2} \right) a^2 \right\}.$$

For a monotone Boolean noise image, $D_i \leq D(a_i)$. Consequently,

$$\sum_{S \neq \emptyset} \omega(|S|) \hat{u}(S)^2 \geq \omega(d)(\mathbb{E}u^2 - m^2) - \sum_i D(a_i). \quad (19)$$

1) *Logarithmic-Sobolev bound via positive smoothing*: For a nonnegative cube function v , define $\text{Ent}(v) = \mathbb{E}[v \log v] - (\mathbb{E}v) \log(\mathbb{E}v)$, with $0 \log 0 = 0$. For any real-valued cube function w , the standard cube logarithmic Sobolev inequality (LSI) [29], [31] is

$$\text{Ent}(w^2) \leq 2\mathbb{E}[w\mathcal{L}w].$$

It holds in every dimension with the same constant; a complete two-point and tensorization proof is in Section A-D. For nonzero, nonnegative w , let $s = \mathbb{E}w^2$ and $b = \mathbb{E}w$. Weight each cube point by w^2/s , and denote expectation under this probability measure by $\mathbb{E}_*$. Jensen gives

$$\mathbb{E}_* \log w = -\mathbb{E}_* \log(1/w) \geq -\log \mathbb{E}_*(1/w) = \log(s/b).$$

Points where $w = 0$ have zero weight. It follows that $\text{Ent}(w^2) = s(2\mathbb{E}_* \log w - \log s) \geq s \log(s/b^2)$. The logarithmic Sobolev bound therefore gives

$$\mathbb{E}[w(\mathcal{L} + 1 - d)w] \geq \frac{s}{2} \log \frac{s}{b^2} + (1 - d)s.$$

The right side is minimized at $s = b^2 e^{2d-3}$, giving

$$\mathbb{E}[w(\mathcal{L} + 1 - d)w] \geq -\frac{1}{2} e^{2d-3} (\mathbb{E}w)^2. \quad (20)$$

The zero function is handled by continuity.

Now average this inequality over ordinary noise smoothings of the coordinate derivative. Let $\mathcal{L}_{-i}$ be the Laplacian on the coordinates other than i , and, for $0 \leq t \leq 1$, set

$$p_i = \frac{u_{i,+} - u_{i,-}}{2}, \quad w_t = T_{\sqrt{t}} p_i.$$

Monotonicity gives $p_i \geq 0$, so $w_t \geq 0$, and every smoothing has the same mean $\mathbb{E}w_t = \rho a_i$. There is an exact identity:

$$D_i = \frac{\gamma}{\gamma + d} \int_0^1 (1 - t^\gamma) \mathbb{E}[w_t(d - 1 - \mathcal{L}_{-i}w_t)] dt.$$

To check it, a Fourier set $S \ni i$ of size s becomes a set of size $s - 1$ after taking the derivative. Its squared coefficient in w_t is $t^{s-1} \hat{u}(S)^2$, and

$$\begin{aligned} & \frac{\gamma(d-s)}{\gamma+d} \int_0^1 (1-t^\gamma) t^{s-1} dt \\ &= \frac{\gamma^2(d-s)}{(\gamma+d)s(\gamma+s)} = \frac{\omega(d) - \omega(s)}{s}. \end{aligned}$$

This is precisely its coefficient in D_i .

Apply (20) to each w_t . The expectation in the integral is at most $e^{2d-3}(\rho a_i)^2/2$. Since $\int_0^1 (1-t^\gamma) dt = \gamma/(\gamma+1)$, we obtain $D_i \leq K_{\text{LS}} \rho^2 a_i^2$.

Remark 8 (Noise averaging). The integral creates exactly the spectral weight defining the deficit. Every field in the average stays nonnegative and has the same mean, so the same logarithmic Sobolev estimate applies throughout.

Samorodnitsky's noisy-function entropy estimates [7] likewise exploit structure preserved under noise. The calculation here uses the standard LSI directly: its positive integral is chosen to reproduce the displayed resolvent weight, so no separate hypercontractive approximation of that weight is required.

2) *Indicator-energy bound*: The unsmoothed derivative q_i^f is an indicator with mean a_i . Its total Fourier energy is a_i ; its constant coefficient contributes a_i^2 . Therefore

$$\sum_{S \ni i} \frac{\hat{f}(S)^2}{|S|} \leq a_i^2 + \frac{1}{2}(a_i - a_i^2) = \frac{a_i + a_i^2}{2}.$$

For degrees $s \geq 2$, the coefficient $[\omega(d) - \omega(s)]\rho^{2s}$ is at most $d_2 \geq 0$. If it is negative this is immediate. Otherwise $2 \leq s < d$, and

$$\omega(d) - \omega(s) = \frac{\gamma^2(d-s)}{(\gamma+d)(\gamma+s)}.$$

On the positive interval, the only s -dependent factor decreases, because

$$\frac{d}{ds} \log \frac{(d-s)\rho^{2s}}{\gamma+s} = -\frac{1}{d-s} + 2 \log \rho - \frac{1}{\gamma+s} < 0.$$

Hence its largest integer value occurs at degree two. Separate the singleton contribution to obtain

$$D_i \leq \frac{d_2}{2}(a_i + a_i^2) + (d_1 - d_2)a_i^2.$$

This proves both bounds in Lemma 7. Finally, summing D_i counts every nonempty set S exactly $|S|$ times against its factor $1/|S|$. The cancellation gives (19).

D. Entropy-Funded Spectral Baseline

The centered angle comparison (11) is valid for every $\ell \geq 31/20$, not just the middle range. We now combine it with the coordinate deficits. Write $\Theta = \arcsin u$ for the angle field and $\theta = \arcsin \rho$ for the dictator's angle, and set

$$p = (\lambda + 1/\alpha)^2, \quad b = \lambda^2(2 + 1/\gamma), \quad C_d = p\omega(d) - b.$$

From this point onward, p denotes this spectral multiplier, not an endpoint value from the earlier edge calculation. The coefficient p comes from the same completed square as before; no new angle approximation or penalty is introduced. Define

$$B = 2\theta^2 + C_d \frac{\phi(\rho)}{L}, \quad k_\Delta = 2\gamma + \frac{C_d}{L}.$$

The information premise bounds the noisy second moment below, with a bound that increases with the lead. When $C_d \geq 0$, that gives an additional positive contribution to the angle energy.

The next lemma lower-bounds the required angle energy after accounting for the coordinate losses. Its two conditions have separate roles: the first permits the second-moment substitution, and the second makes the remaining bias contribution non-negative. They are proved for the chosen channel parameters in the tail appendix, rather than assumed as properties of the Boolean rule.

Lemma 9 (An entropy-funded spectral baseline). *Let $u = T_\rho f$ for a monotone Boolean function f , with $\ell = \text{atanh } \rho \geq 31/20$ and the canonical penalty $\gamma = \lambda\alpha$. Suppose*

$$C_d \geq 0, \quad \gamma - b + C_d \left(\frac{1}{2L} - 1 \right) \geq 0. \quad (21)$$

Then

$$\mathbb{E}[\Theta \mathcal{L} \Theta] \geq B - p \sum_i D(a_i) + k_\Delta \Delta_f(\rho). \quad (22)$$

Proof. The identity $\gamma = \lambda\alpha$ gives, for every degree $s \geq 0$,

$$\lambda^2 \left[\gamma - \frac{(\gamma + 1)^2}{\gamma + s} \right] = p\omega(s) - b.$$

Insert it into (11), including the constant mode, and then use Lemma 7:

$$\begin{aligned} \mathbb{E}[\Theta \mathcal{L} \Theta] &\geq 2\theta^2 + C_d \text{Var}(u) - p \sum_i D(a_i) \\ &\quad + 2\gamma\phi(m) - bm^2 + 2\gamma\Delta_f(\rho). \end{aligned}$$

The entropy series bounds $\phi(t) \leq Lt^2$ on $[-1, 1]$:

$$\phi(t) = \sum_{j \geq 1} \frac{t^{2j}}{2j(2j-1)} \leq t^2 \sum_{j \geq 1} \frac{1}{2j(2j-1)} = Lt^2.$$

As $\mathbb{E}\phi(u) = \phi(\rho) + \phi(m) + \Delta_f(\rho)$, this yields

$$\text{Var}(u) \geq \frac{\phi(\rho) + \phi(m) + \Delta_f(\rho)}{L} - m^2.$$

Its coefficient C_d is nonnegative, so substitution preserves the lower bound. The channel and lead terms are B and $k_\Delta \Delta_f(\rho)$. The remaining mean term is exactly

$$\begin{aligned} &\left(2\gamma + \frac{C_d}{L} \right) \phi(m) - (C_d + b)m^2 \\ &\geq \left[\gamma - b + C_d \left(\frac{1}{2L} - 1 \right) \right] m^2 \geq 0, \end{aligned}$$

using $\phi(m) \geq m^2/2$. Dropping only this nonnegative term proves the claim. No sign assumption on Δ_f was needed. $\square$

The mean was never set to zero. Conditions (21) give the signs needed to use the second-moment bound and discard the mean remainder. The tail appendix proves them for the canonical penalty and the displayed degree.

E. Yield Comparison and Growth

For $0 < v \leq \ell_\rho$, set $a = F(\ell_\rho)/F(v)$ and define

$$\text{yield}(v) = \frac{\rho c(v)}{v} + \frac{pD(a)}{av}.$$

This *allowed yield* is angle contribution plus Fourier loss, per unit of edge weight av . The *required yield* B/ℓ_ρ is the channel

baseline per unit dictator weight. These are comparison bounds, not coordinate mutual informations.

The next lemma shows that the coordinate losses cannot consume the extra energy required by an information lead when every coordinate's allowed yield is below the required yield. Its proof combines two accounts of the same growth: angle energy plus correction and the total edge-weight budget.

Lemma 10 (The yield comparison forces faster growth). *Let $u = T_\rho f$ for a monotone Boolean f , with $\ell = \text{atanh } \rho \geq 31/20$ and the canonical penalty. Assume $\Delta_f(\rho) = I_f(\rho) - \phi(\rho) \geq 0$ and the two conditions (21). If*

$$\frac{B}{\ell_\rho} \geq \rho, \quad \frac{B}{\ell_\rho} \geq \text{yield}(\ell_i) \quad \text{for every active coordinate } i, \quad (23)$$

then

$$\Delta'_f(\rho) \geq \frac{k_\Delta \ell_\rho}{B} \Delta_f(\rho). \quad (24)$$

In particular, a positive information lead has positive growth.

Proof. The lead assumption gives $\mathbb{E}h(u) = h(\rho) - \phi(m) - \Delta_f(\rho) \leq h(\rho)$, so the edge lemma applies. Its correction and the spectral baseline concern the same coordinate weights. Combining them gives

$$\begin{aligned} A(u) &\geq \underbrace{\mathbb{E}[\Theta \mathcal{L} \Theta]}_{\text{angle energy}} + \underbrace{\rho W - \rho \sum_i a_i c(\ell_i)}_{\text{edge correction}} \\ &\geq \underbrace{B + k_\Delta \Delta_f(\rho)}_{\text{required baseline and lead credit}} + \sum_i a_i \ell_i [\rho - \text{yield}(\ell_i)] \\ &\geq B + k_\Delta \Delta_f(\rho) + \left(\rho - \frac{B}{\ell_\rho} \right) W. \end{aligned}$$

The first line is Lemma 5; the second inserts Lemma 9 and groups each coordinate's two costs into its yield. The last line uses the scalar yield comparison. Thus the many coordinate losses now depend only on the single total weight W . Now use the other edge bound, $W \leq A(u)/\rho$. Its coefficient in the last line is nonpositive, so substitution gives The resulting growth budget is

$$\frac{B}{\rho \ell_\rho} A(u) \geq B + k_\Delta \Delta_f(\rho). \quad (25)$$

Since $B/\ell_\rho \geq \rho > 0$, dividing and using $\Delta'_f = A(u)/\rho - \ell_\rho$ proves (24). The lead supplies the positive credit that forces its own growth. Lemma 13 establishes the scalar comparisons needed here. $\square$

V. PROOF OF THE MAIN THEOREM

We now state the conclusion in the variables of the original problem. Recall that $I_f(\rho)$ is the retained bit's mutual information, $\phi(\rho)$ is the dictator's information, and a prime means $d/d\rho$ with the function f fixed.

A. Direct Bounds

These bounds already settle two classes of possible counterexamples. They are logically separate from the spectral comparison.

Proposition 11 (Small correlation). *For every Boolean function f , of any mean, and every $0 \leq \rho \leq \rho_*$,*

$$I_f(\rho) \leq \phi(\rho).$$

Proof. Appendix D proves this from an elementary quantitative Fourier estimate, an entropy minorant, and entropy contraction. The supporting estimates are proved there and in Appendix A. $\square$

Proposition 12 (A sufficiently influential coordinate). *Let f be Boolean, $0 < \rho < 1$, and let $a = |\hat{f}(\{i\})|$ for some coordinate i . If*

$$(1 - \rho^2)h(\rho a) \geq (1 - \rho^2 a)h(\rho), \quad (26)$$

then $I_f(\rho) \leq \phi(\rho)$.

Proof. This is the influential-coordinate part of the local/Fourier division used by Yu [11, Section 4.2]. Appendix A proves the displayed criterion with its mean remainder retained. For monotone f , a is the coordinate's influence; dictators have $a = 1$ and give equality. $\square$

B. Global Growth Comparison

A violating function must have $\rho > \rho_*$, and every one of its coordinates must fail (26). Its coordinate influences therefore lie in the domain of the yield comparison below.

Lemma 13 (An advantage must be growing). *For every monotone Boolean function f and every $0 < \rho < 1$,*

$$I_f(\rho) > \phi(\rho) \implies I'_f(\rho) > \phi'(\rho).$$

In words: wherever a rule beats the dictator, its lead is increasing.

Proof. Proposition 11 excludes $\ell \leq 31/20$. On $31/20 \leq \ell \leq 7/2$, Section IV proves the claim by the entropy and log-sum argument. For $\ell \geq 7/2$, Proposition 12 excludes the influential-coordinate case. On all remaining coordinates, Appendix F proves, with the same canonical penalty and $d = \ell/2 + 5/4$, both conditions in (21) and

$$B/\ell \geq \max\{\rho, \text{yield}(v)\} \quad (27)$$

for every retained height v . The continuum of heights reduces to the half-height join and the cutoff by Appendix E. Three half-line comparisons prove the bounds, with no bounded cutoff ranges. Lemma 10 now gives $A(u) > \rho\ell$. Since $A(u) = \rho I'_f(\rho)$ and $\phi'(\rho) = \ell$, this is precisely the conclusion. $\square$

Proof of Theorem 2. By Lemma 3, it suffices to consider monotone functions. Fix one and write $\Delta_f = I_f - \phi$. At the noiseless endpoint, $\Delta_f(1) = -\phi(m) \leq 0$.

Suppose Δ_f is positive somewhere. Choose any point ρ_0 in a maximal open interval on which it is positive, and call that interval's right endpoint $b \leq 1$. Lemma 13 makes Δ_f strictly increasing on this interval. Consequently $\lim_{\rho \uparrow b} \Delta_f(\rho) \geq \Delta_f(\rho_0) > 0$. Continuity gives $\Delta_f(b) = 0$ if $b < 1$, or $\Delta_f(b) \leq 0$ if $b = 1$. Either case is a contradiction. $\square$

VI. VECTOR-VALUED QUANTIZATION AND A THRESHOLD AT EIGHT BITS

Theorem 2 motivates the vector-valued question of whether a k -bit summary is optimized by retaining k input coordinates. The answer is negative in general. This section gives explicit counterexamples, describes their coding-theoretic mechanism, and isolates the range that remains open.

The outcome is unexpectedly sharp. Shortened Hamming quantizers violate the coordinate benchmark for $k = 8, 9, 10$, and hence, by adding independent coordinate outputs, for every $k \geq 8$. The same construction misses the benchmark at $k = 7$ by only $1/64$ in the high-noise coefficient. A broad collection of searches below eight bits also failed to produce a violation. This leads us to conjecture that the coordinate projection is optimal for $2 \leq k \leq 7$.

Throughout this section, entropies and mutual informations are measured in bits. Let

$$\begin{aligned} X &\sim \text{Unif}(\mathbb{F}_2^n), & Y &= X \oplus Z, \\ Z_i &\stackrel{\text{iid}}{\sim} \text{Bernoulli}(p), & 0 &\leq p \leq \frac{1}{2}. \end{aligned}$$

and write

$$h_2(p) = -p \log_2 p - (1-p) \log_2 (1-p), \quad C(p) = 1 - h_2(p).$$

A. Problem Formulation and Known Results

For $n \geq k$, define

$$M_k(n, p) := \max_{f: \mathbb{F}_2^n \rightarrow \mathbb{F}_2^k} I(f(X); Y). \quad (28)$$

The k -coordinate projection

$$D_k(x) = (x_1, \dots, x_k)$$

satisfies $I(D_k(X); Y) = kC(p)$. We call D_k , and any version obtained by permuting input coordinates and relabeling its 2^k outputs, a *dictator tuple*. The most direct vector extension of the Courtade–Kumar assertion is therefore

$$M_k(n, p) \stackrel{?}{\leq} kC(p) \quad \text{for every } n \geq k. \quad (29)$$

There are three useful landmarks in the literature.

- 1) For $k = 1$, inequality (29) is the original Courtade–Kumar conjecture [1], proved in this paper.
- 2) Huleihel and Ordentlich proved the complementary endpoint: for every $f: \mathbb{F}_2^n \rightarrow \mathbb{F}_2^{n-1}$,

$$I(f(X); Y) \leq (n-1)C(p).$$

Their theorem also extends from the BSC to every binary-input memoryless output-symmetric channel [27]. Notice that this is an assertion with $k = n-1$, not an assertion for a fixed k and arbitrary n ; hence it does not conflict with the fixed- k counterexamples below.

- 3) Chandar and Tchamkerten formulated the general optimization problem (28) and characterized its positive-rate asymptotics [26]. If $k/n \rightarrow R \in (0, 1)$, then

$$\begin{aligned} \frac{M_k(n, p)}{n} &\longrightarrow 1 - h_2(p \star h_2^{-1}(1-R)), \\ p \star q &= p(1-q) + (1-p)q. \end{aligned} \quad (30)$$

This limit can exceed $RC(p)$, so the vector assertion cannot hold at every positive rate. At finite blocklength, they exhibited violations from the perfect [15, 11, 3] Hamming code and the [23, 12, 7] Golay code. They also observed that the [7, 4, 3] Hamming code does not violate the bound and explicitly asked for constructions with $k < 11$.

One elementary point prevents a possible misunderstanding: a linear *code* can produce a nonlinear *quantizer*.

Proposition 14 (Affine maps obey the coordinate bound). *If $f(x) = Ax + b$ is affine, with output rank $r \leq k$, then*

$$I(f(X); Y) \leq rC(p) \leq kC(p).$$

Proof. It is enough to treat a rank- r linear map. Since X is uniform, AX is uniform on $\mathbb{F}_2^r$. Moreover, $Z = X \oplus Y$ is independent of Y , and hence

$$I(AX; Y) = r - H(AZ).$$

Choose r columns of A that form an invertible submatrix and call their index set J . Conditioning on Z_{J^c} , the variable AZ is a translate of an invertible image of Z_J . Consequently,

$$H(AZ) \geq H(AZ | Z_{J^c}) = H(Z_J) = rh_2(p),$$

which proves the claim. $\square$

Thus every counterexample must use genuine nonlinearity. The constructions below use linear codebooks, but the nearest-codeword map into those codebooks is nonlinear.

B. Coset Quantizers and the High-Noise Expansion

Let $C \leq \mathbb{F}_2^n$ be a binary linear code of dimension k , with full-rank parity-check matrix $H \in \mathbb{F}_2^{(n-k) \times n}$. For each syndrome $a \in \mathbb{F}_2^{n-k}$, choose a minimum-weight coset leader e_a satisfying $He_a = a$, with $e_0 = 0$, and define

$$Q(x) = x \oplus e_{Hx} \in C. \quad (31)$$

After composing Q with any bijection from C to $\mathbb{F}_2^k$, this is a k -bit quantizer. The choice among tied minimum-weight leaders is part of the construction.

If X is uniform, the change of variables

$$X = Q(X) \oplus E, \quad E = e_{HX},$$

makes $Q(X)$ uniform on C , makes E uniform on the set of coset leaders, and makes the two variables independent. Since Y is uniform on $\mathbb{F}_2^n$, translation invariance gives the exact identity

$$I(Q(X); Y) = n - H(E \oplus Z). \quad (32)$$

This reduces the entire information calculation to the entropy of a noisy coset leader.

Set $\rho = 1 - 2p$ and define

$$B(E) := \sum_{j=1}^n (\mathbb{E}[(-1)^{E_j}])^2. \quad (33)$$

For a set S , the Walsh coefficient of $E \oplus Z$ is

$$\rho^{|S|} \mathbb{E}[(-1)^{\sum_{j \in S} E_j}].$$

Expanding relative entropy from the uniform distribution at $\rho = 0$ therefore yields

$$I(Q(X); Y) = \frac{\rho^2}{2 \ln 2} B(E) + O(\rho^4), \quad (34)$$

$$kC(p) = \frac{\rho^2}{2 \ln 2} k + O(\rho^4). \quad (35)$$

In particular,

$$B(E) > k \implies I(Q(X); Y) > kC(p) \quad (36)$$

for all $p < 1/2$ sufficiently close to $1/2$.

This exact second-order test is the main design criterion used below.

C. Shortened-Hamming Counterexamples for $k = 8, 9, 10$

Identify the nonzero vectors of $\mathbb{F}_2^4$ with the integers $1, \dots, 15$ through their binary expansions. For $s \in \{1, 2, 3\}$, let H_s have as its columns

$$\{s+1, s+2, \dots, 15\}.$$

Thus $C_s = \ker H_s$ is the shortened Hamming code

$$C_s = [15-s, 11-s, 3], \quad n = 15-s, \quad k = 11-s. \quad (37)$$

The retained syndrome 8 will be an anchor. Choose the following minimum-weight leader for every syndrome $a \in \mathbb{F}_2^4$:

$$e_a = \begin{cases} 0, & a = 0, \\ e_a, & a \in \{s+1, \dots, 15\}, \\ e_8 \oplus e_{8 \oplus a}, & a \in \{1, \dots, s\}. \end{cases} \quad (38)$$

Here e_a denotes the unit vector in the coordinate whose parity-check column is a . Both 8 and $8 \oplus a$ are retained. The last line has syndrome a , and it is minimum-weight because the column a itself was removed. The missing syndromes therefore use weight-two leaders that form a star around the common anchor 8.

The leader set consists of one word of weight zero, n words of weight one, and s words of weight two. More importantly, the anchor occurs in $s+1$ leaders, each of the s leaves occurs in two leaders, and each of the remaining $14-2s$ coordinates occurs in one leader. Hence

$$\begin{aligned} B_s &= \left(1 - \frac{s+1}{8}\right)^2 + s \left(1 - \frac{2}{8}\right)^2 \\ &\quad + (14-2s) \left(1 - \frac{1}{8}\right)^2 \\ &= \frac{s^2 - 76s + 735}{64}, \end{aligned} \quad (39)$$

$$B_s - k = \frac{s^2 - 12s + 31}{64}. \quad (40)$$

Proposition 15 (Counterexamples below eleven bits). *For each $k \in \{8, 9, 10\}$, there are an $n = k + 4$, a quantizer $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^k$, and an open interval of crossover probabilities ending at $1/2$ on which*

$$I(f(X); Y) > kC(p).$$

Proof. Use the quantizer (31) with the code and leaders in (37)–(38). For $s = 1, 2, 3$, corresponding respectively to $k = 10, 9, 8$,

$$B_s - k \in \left\{ \frac{5}{16}, \frac{11}{64}, \frac{1}{16} \right\} > 0.$$

The conclusion follows from the high-noise expansion (36). $\square$

Proposition 16 (Failure for every $k \geq 8$). *For every integer $k \geq 8$, inequality (29) fails for some input dimension and some $p \in (0, 1/2)$.*

Proof. Start with the 8-bit counterexample f_8 . For $t = k - 8$, append t fresh independent input coordinates and define

$$F(x, u) = (f_8(x), u), \quad u \in \mathbb{F}_2^t.$$

The two channel blocks are independent, so

$$I(F(X, U); Y_X, Y_U) = I(f_8(X); Y_X) + tC(p) > kC(p).$$

$\square$

The proof is analytic and uses only the sign of the exact rational number $B_s - k$. Direct entropy evaluation gives a more quantitative picture. For $y \in \mathbb{F}_2^n$, we computed

$$\Pr(E \oplus Z = y) = \frac{1}{16} \sum_{a \in \mathbb{F}_2^4} p^{d_H(y, e_a)} (1 - p)^{n - d_H(y, e_a)} \quad (41)$$

and substituted the resulting distribution in (32). In Table I, the leader enumerator is $\sum_a z^{\text{wt}(e_a)}$. The positive intervals and maxima are numerical; the code parameters, leader enumerators, and $B_s - k$ are exact.

Here

$$\Delta_k(p) = I(f(X); Y) - kC(p).$$

The maximizing crossover probabilities are approximately 0.199318, 0.248809, and 0.332782, respectively. Simple interior witness points are

$$\begin{aligned} \Delta_{10}(0.2)/10 &= 0.0044842878, \\ \Delta_9(0.25)/9 &= 0.0018330159, \\ \Delta_8(1/3)/8 &= 0.0003212940. \end{aligned}$$

Direct convolution and an independent Walsh-transform calculation agreed to within 10^{-14} at the tested crossover probabilities.

Remark 17 (Order of shortening and decoding). The successful operation is to remove columns from the Hamming parity-check matrix and then choose a new minimum-weight syndrome table. Deleting output bits after decoding with the full $[15, 11, 3]$ Hamming quantizer is a different operation and did not reproduce these gains. The shared anchor in (38) is also essential: it aligns the coordinate biases that enter $B(E)$.

D. Searches Below Eight Bits

The star construction itself explains why eight is a natural boundary. For $s = 4, 5, 6$, corresponding to $k = 7, 6, 5$, formula (40) gives The seven-bit case is strikingly close: its leading high-noise term misses the dictator tuple by only $1/64$. A dense numerical evaluation found no positive gap on $0 < p < 1/2$; for example, the per-bit deficits at $p = 0.1, 0.2, 0.4$ are respectively

$$0.0115176, \quad 0.00410469, \quad 0.000105113.$$

We then tested several ways in which the Hamming near-miss might be improved. Table III records the scope of each computation and the best high-noise statistic found. A negative-curve outcome means that the full mutual-information curve was also evaluated on a dense grid and at its numerical stationary candidates; it is not an analytic impossibility proof.

We also tested repeated parity-check columns, nonminimum coset representatives, the extended $[16, 7, 6]$ BCH construction, the $[15, 5, 7]$ BCH code, and $\text{RM}(1, 4) = [16, 5, 8]$. Repeated-column searches occasionally reached $B = 7$, but only through a degeneracy whose entire information curve equals that of seven independent dictators; no strict gain appeared. Multistart searches over arbitrary coset representatives likewise reached equality but not a violation.

These computations support a structural claim about the code-based mechanism, not a theorem about all quantizers. In particular, the exact exhaustion of Hamming tie tables certifies the best B only within that family. A quantizer with a smaller high-noise coefficient could still, in principle, cross the benchmark at an intermediate p .

E. Random Coding at $n = 15$, $k = 7$

The failed structured searches raise a natural question: is an unstructured codebook more likely to find the missing gain? We tested the finite random ensemble suggested by the rate-distortion picture. Choose a codebook $\mathcal{C}$ uniformly without replacement from all $M = 2^7$ -subsets of $\mathbb{F}_2^{15}$, map each input to a nearest codeword, and break each tie independently and uniformly, once and for all.

Some geometric statistics of this ensemble are exact. Let

$$N = 2^{15}, \quad S_d = \binom{15}{d}, \quad V_d = \sum_{i=0}^d S_i.$$

For a fixed source word, let D be its distance to the random codebook and T the number of codewords at that minimum distance. Then

$$\Pr(D > t) = \frac{\binom{N-V_t}{M}}{\binom{N}{M}}, \quad (42)$$

$$\Pr(D = d, T = m) = \frac{\binom{S_d}{m} \binom{N-V_d}{M-m}}{\binom{N}{M}}. \quad (43)$$

For $n = 15$ and $M = 128$, these formulas give

$$\begin{aligned} \mathbb{E}D &= 2.6608532, & \frac{\mathbb{E}D}{15} &= 0.1773902, \\ \Pr(T > 1) &= 0.4945889, & \mathbb{E}T &= 2.1208972. \end{aligned}$$

Thus almost half of the source words have a nearest-neighbor tie.

For an arbitrary quantizer $U = Q(X)$, not necessarily a coset quantizer, the high-noise coefficient is

$$B(Q) = \sum_{j=1}^n \sum_u \Pr(U = u) (\mathbb{E}[(-1)^{X_j} \mid U = u])^2, \quad (44)$$

and

$$I(U; Y) = \frac{\rho^2}{2 \ln 2} B(Q) + O(\rho^4). \quad (45)$$

TABLE I

SHORTENED-HAMMING COUNTEREXAMPLES. THE CODE PARAMETERS, LEADER ENUMERATORS, AND HIGH-NOISE EXCESSES ARE EXACT; THE INTERVALS AND MAXIMA ARE OBTAINED BY DIRECT ENTROPY EVALUATION.

k	code	leader enumerator	$B_s - k$	numerical p -interval	$\max_p \Delta_k(p)/k$
10	[14, 10, 3]	$1 + 14z + z^2$	5/16	(0.088658, 1/2)	4.48440×10^{-3}
9	[13, 9, 3]	$1 + 13z + 2z^2$	11/64	(0.152276, 1/2)	1.83320×10^{-3}
8	[12, 8, 3]	$1 + 12z + 3z^2$	1/16	(0.265563, 1/2)	3.21308×10^{-4}

TABLE II

HIGH-NOISE COEFFICIENTS OF THE SHORTENED-HAMMING FAMILY BELOW EIGHT OUTPUT BITS.

k	n	leader enumerator	B_s	$B_s - k$
7	11	$1 + 11z + 4z^2$	447/64	-1/64
6	10	$1 + 10z + 5z^2$	95/16	-1/16
5	9	$1 + 9z + 6z^2$	315/64	-5/64

We enumerated the full cube for 128 independently sampled codebooks and evaluated every reported mutual information exactly for that sampled quantizer. Thus the only Monte Carlo error is across codebooks, not within a codebook. The comparison below includes a structured code of the same length and size, and the shortened-Hamming near-miss of a different length. For the BCH code, the displayed syndrome table was optimized over its minimum-distance ties.

Across the 128 random codebooks, $B(Q)$ ranged only from 6.342219 to 6.423068, and no sampled codebook beat the coordinate benchmark at any tested crossover probability. The random Voronoi cells were also appreciably unbalanced: their mean cell-size standard deviation was 47.31 around the ideal size 256. This explains the loss $H(U) < 7$ already at $p = 0$. The sample standard deviation of $B(Q)$ was 0.01856, giving the normal-approximation 95% interval [6.3762, 6.3826] for its ensemble mean.

Because ties are common, we separately optimized their assignments for the high-noise statistic. On 64 additional codebooks, a multistart coordinate-ascent heuristic increased the mean B from 6.3809 to 6.6954, with a best value 6.71365. This is a material improvement, but it remains well below both 7 and the shortened-Hamming value 6.984375. Since the tie optimization is heuristic, it is evidence rather than an ensemble upper bound.

The experiment suggests that short-block structure matters in this problem. Translation-equivariant code quantizers have equal-size cells and can align their coordinate biases; a generic random Voronoi partition has neither feature. This conclusion is qualitatively consonant with the recent finite-block covering results of Riasat and MahdaviFar, where structured Reed–Solomon-based constructions often outperform random codebooks in average Hamming covering radius [28]. The two statements should not be conflated: their objective is average nearest-codeword distance, whereas ours is BSC mutual information. The agreement is a shared finite-block phenomenon, not a formal implication.

F. A Seven-Bit Threshold Conjecture

The validity of the coordinate bound is monotone in the number of retained bits in a useful sense. If a k -bit counterexample exists, appending one fresh dictator gives a $(k + 1)$ -bit counterexample. Conversely, if the bound is valid for K bits in every dimension, then it is valid for every $k < K$: append $K - k$ fresh dictators to a hypothetical k -bit counterexample. Thus the set of fixed output sizes for which the bound is universally true must be an initial segment of the positive integers.

The constructions above show that this initial segment ends no later than seven. We conjecture that it ends exactly there.

Conjecture 18 (Seven-bit threshold). *For every $2 \leq k \leq 7$, every $n \geq k$, every $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^k$, and every $0 \leq p \leq 1/2$,*

$$I(f(X); Y) \leq k(1 - h_2(p)). \quad (46)$$

Equality is attained by a k -coordinate projection.

By the padding observation, it would be enough to prove Conjecture 18 for $k = 7$. Several pieces of evidence point in the same direction.

- 1) *The Hamming family changes sign at exactly the proposed threshold.* Its exact excess coefficient is $(s^2 - 12s + 31)/64$: it is 1/16 at $k = 8$ and $-1/64$ at $k = 7$. This is not a numerical accident or a small interior- p margin.
- 2) *Linear maps cannot be counterexamples.* By Proposition 14, a violation must combine nonlinear cells with enough symmetry to preserve almost all k output bits. Our unsuccessful postprocessing experiments illustrate the tension: coarsening an eight-bit counterexample destroys the bias alignment much faster than it reduces the benchmark.
- 3) *The first high-noise target is concrete.* A proof of

$$B(Q) \leq 7 \quad \text{for every } Q : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^7 \quad (47)$$

would settle the sharp second-order comparison at $p = 1/2$. Every seven-bit family we tested satisfies this inequality, with the shortened Hamming construction coming closest among the nontrivial examples.

- 4) *Random coding points away from a hidden generic counterexample.* Random nearest-neighbor quantizers are substantially worse than the structured near-misses, even after aggressive tie optimization. If the conjecture is false, the counterexample is likely to require a rather special multiway partition rather than merely a good covering code.

TABLE III
COMPUTATIONAL SEARCHES FOR SEVEN-BIT CODE QUANTIZERS. A “NEGATIVE CURVE” RECORDS DENSE NUMERICAL EVALUATION AND STATIONARY-POINT CHECKS, NOT AN IMPOSSIBILITY THEOREM FOR THE CORRESPONDING FAMILY.

Family	Search performed	best B	outcome
Shortened Hamming, $k = 7$	All $\binom{15}{4} = 1365$ shortenings and all 451,920 minimum-distance tie tables	447/64	negative curve
Primitive BCH [15, 7, 5]	All 1941 puncturings deleting 0, ..., 4 coordinates, with multistart optimization of minimum-distance ties	447/64	best case returns the Hamming near-miss
First- and second-order Reed–Muller family	All 651 seven-dimensional codes $\text{RM}(1, 4) \subset C \subset \text{RM}(2, 4)$	6.850952	negative curves
Shortened binary Golay	203 sampled shortenings to dimension seven, with tie optimization	6.895	negative curves
Random parity-check matrices	345,000 distinct-column searches with redundancies 5, 6, 7, 8	447/64	no improvement over Hamming
Linear quotients of the $k = 8$ counterexample	All 255 rank-seven quotients of the [12, 8, 3] quantizer	6.28125	negative curves

TABLE IV
RANDOM AND STRUCTURED SEVEN-BIT QUANTIZERS. MUTUAL INFORMATION IS EVALUATED EXACTLY FOR EACH SAMPLED QUANTIZER; ONLY THE AVERAGING OVER RANDOM CODEBOOKS INTRODUCES MONTE CARLO UNCERTAINTY.

Quantity	random (15, 2 ⁷), mean	BCH [15, 7, 5]	Hamming [11, 7, 3]
$H(U)$	6.975169	7	7
$B(Q)$	6.379431	6.850952	6.984375
$\Delta_7(0.1)/7$	−0.0505444	−0.0294541	−0.0115176
$\Delta_7(0.4)/7$	−0.00262873	−0.000697266	−0.000105113

There is also a clear reason the scalar theorem does not immediately iterate. Writing $U = (U_1, \dots, U_k)$, the chain rule gives

$$I(U; Y) = \sum_{i=1}^k I(U_i; Y \mid U_1, \dots, U_{i-1}),$$

but after conditioning on the previous output bits, X is uniform only on a cell of the induced partition, not on the full cube. The scalar Courtade–Kumar theorem therefore does not apply to the conditional terms. A proof of the seven-bit conjecture will likely need either a conditional version of the scalar inequality or a genuinely multiway entropy-flow argument.

Remark 19 (A possible route for $k = 7$). The conjecture for all $2 \leq k \leq 7$ reduces to one endpoint, $k = 7$. A plausible program is to prove the high-noise inequality (47), identify its equality cases, obtain a multiway low-noise boundary inequality for balanced 128-cell partitions, and then bridge the two channel regimes. The 1/64 Hamming near-miss provides a useful stress test for every proposed estimate.

VII. CONCLUSION

We proved that a coordinate is the most informative Boolean function of a uniform binary vector observed through a BSC. The proof combines monotone sorting, entropy flow, an entropy-constrained edge envelope, and spectral information carried by pivotal sets. Its key global feature is the growth implication: a positive advantage over a dictator would have to increase with channel correlation and therefore cannot be reconciled with the noiseless endpoint.

The vector-valued problem exhibits a different behavior. Shortened Hamming quantizers violate the coordinate benchmark for every fixed $k \geq 8$, whereas extensive structured and

random searches produced no violation for $k \leq 7$. This leaves a concrete threshold problem. A particularly useful first step would be to prove the sharp high-noise bound $B(Q) \leq 7$ for every seven-bit quantizer and characterize its equality cases. A complete result will likely require either a conditional form of the scalar theorem or a genuinely multiway entropy-flow inequality, since the ordinary chain rule does not preserve the uniform cube measure after conditioning.

GENERATIVE AI DISCLOSURE

The proof of the single-bit result, including its detailed mathematical derivations, was generated by generative AI systems; the authors’ role was high-level direction, feedback, and final review. The multibit extension was entirely human-directed: the authors formulated the problem, selected the code-based constructions and computational questions, and interpreted the evidence and conjectures. Generative AI systems assisted with developing and checking proofs, implementing and verifying computations, and refining the exposition. The authors reviewed the complete manuscript and accept full responsibility for its mathematical claims, citations, and final text.

APPENDIX A

SCALAR EDGE AND ENTROPY INEQUALITIES

This appendix proves the scalar and edge facts used in Section II. Write

$$\begin{aligned} L &= \log 2, & h(r) &= L - \phi(r), \\ g(r) &= \text{atanh } r, & F(v) &= \frac{h(\tanh v)}{\tanh v}. \end{aligned}$$

All these facts are independent of dimension.

A. Convex Lower Bound for Centered Edges

For edge values $p, q \in (-1, 1)$, define

$$\begin{aligned} d_{\text{edge}} &= \frac{|p - q|}{2}, & H_e &= \frac{h(p) + h(q)}{2}, \\ a(p, q) &= \frac{(p - q)(g(p) - g(q))}{4}, \\ j(p, q) &= \frac{(\arcsin p - \arcsin q)^2}{4}. \end{aligned}$$

The edge contributes $a(p, q)$ to the action and $j(p, q)$ to the angle energy. We abbreviate these edge functions as a, j within this subsection. The subscript in d_{edge} names its role: edge half-width. Let F^{-1} denote the inverse function of F . For a positive scalar k , first define

$$\theta_{\text{edge}}(k) = \arcsin(\tanh k), \quad \text{corr}(F(k)) = k - \frac{\theta_{\text{edge}}(k)^2}{\tanh k}.$$

Here corr is short for the retained edge correction. We will prove

$$\begin{aligned} a(p, q) &\geq d_{\text{edge}} F^{-1}(H_e/d_{\text{edge}}), \\ a(p, q) - j(p, q) &\geq d_{\text{edge}} \text{corr}(H_e/d_{\text{edge}}). \end{aligned} \quad (\text{S2})$$

The right-hand sides are convex functions of (d_{edge}, H_e) and decrease with H_e . This permits averaging over all edges in a coordinate. At $d_{\text{edge}} = 0$, both expressions are defined by their limiting value zero.

1) *Centering property*: For any real number r , define $\sinh r = (e^r - e^{-r})/2$ and $\cosh r = (e^r + e^{-r})/2$. For $r \neq 0$, $\coth r = \cosh r / \sinh r$. In this edge calculation v is a real center and $k > 0$ a half-width, specified by $p = \tanh(v + k)$, $q = \tanh(v - k)$. Assume $p > q$. Then $a = d_{\text{edge}} k$. Set $M = \cosh(2v)$ and $C_k = \cosh(2k)$. Direct calculation gives

$$\begin{aligned} d_{\text{edge}} &= \frac{\sinh(2k)}{M + C_k}, \\ H_e &= \frac{1}{2} \log[2(M + C_k)] - \frac{v \sinh(2v) + k \sinh(2k)}{M + C_k}. \end{aligned}$$

To differentiate with respect to the center, first consider $v > 0$ and set $x = 2v$. The derivative of the numerator of H_e/d_{edge} has the sign of

$$\log[2(\cosh x + C_k)] - x \coth x.$$

It is nonnegative because, for $C_k \geq 1$,

$$\begin{aligned} \log[2(\cosh x + C_k)] &\geq x + 2 \log(1 + e^{-x}) \\ &\geq x + \frac{2}{e^x + 1} \\ &\geq x + \frac{2x}{e^{2x} - 1} = x \coth x. \end{aligned}$$

The last two steps use $\log(1 + y) \geq y/(1 + y)$ for $y \geq 0$ and $e^x - 1 \geq x$. Evenness handles $v < 0$. Hence

$$H_e/d_{\text{edge}} \geq F(k). \quad (\text{S3})$$

The function $\arctan x$ is the angle between $-\pi/2$ and $\pi/2$ whose tangent is x . For the angle term, let $b_{\text{edge}} = \sinh k$

and $x = b_{\text{edge}}/\cosh v$. This abbreviation is used only in the following half-angle calculation. The half-angle identity gives

$$\begin{aligned} \arcsin p - \arcsin q &= 2 \arctan x, \\ \frac{j}{d_{\text{edge}}} &= \frac{b_{\text{edge}}}{\sqrt{1 + b_{\text{edge}}^2}} (1 + x^2) \left(\frac{\arctan x}{x} \right)^2. \end{aligned}$$

The last factor increases with x : its logarithmic derivative is $2(x - \arctan x)/[x(1 + x^2) \arctan x] > 0$. Since $x \leq b_{\text{edge}}$, it follows that

$$j/d_{\text{edge}} \leq \theta_{\text{edge}}(k)^2 / \tanh k. \quad (\text{S4})$$

2) *Averaging by convexity*: Put $\ell = \log(2 \cosh k)$. Differentiation gives

$$F'(k) = -\frac{\ell}{\sinh^2 k}, \quad (F^{-1})'(F(k)) = -\frac{\sinh^2 k}{\ell}.$$

The positive ratio on the right increases with k , since $2\ell > \tanh^2 k$. Thus F^{-1} is decreasing and convex.

Similarly,

$$\begin{aligned} \frac{d}{dk} \left(k - \frac{\theta_{\text{edge}}(k)^2}{\tanh k} \right) &= \left(1 - \frac{\theta_{\text{edge}}(k)}{\sinh k} \right)^2, \\ \text{corr}'(F(k)) &= -\frac{(\sinh k - \theta_{\text{edge}}(k))^2}{\ell}. \end{aligned}$$

The positive ratio in the last expression increases with k : its derivative has the sign of $2\ell \sinh k - (\sinh k - \theta_{\text{edge}}(k))^2 > 0$. Therefore corr is nonnegative, decreasing, and convex.

Combining these facts with (S3)–(S4) proves (S2). Let ψ be either of the two functions F^{-1} or corr. The function of two variables $d_{\text{edge}} \psi(H_e/d_{\text{edge}})$, called its *perspective*, is convex, increases with d_{edge} , and decreases with H_e . Jensen may therefore be applied to (S2).

For a monotone field $u = T_p f$, a coordinate i with $a_i > 0$ satisfies $\mathbb{E} d_{\text{edge}} = \rho a_i$ and $\mathbb{E} H_e = \mathbb{E} h(u)$. For any upper bound $H_{\text{budget}} \geq \mathbb{E} h(u) > 0$, Jensen and decrease in entropy give

$$\begin{aligned} \mathbb{E} a &\geq \rho a_i v_i, \\ \mathbb{E}(a - j) &\geq \rho a_i [v_i - c(v_i)], \\ F(v_i) &= H_{\text{budget}}/(\rho a_i). \end{aligned}$$

Summing proves Lemma 5. Its two applications simply supply different entropy budgets.

B. Monotonicity of the Entropy Profile

Lemma 20 (Profile monotonicity). *The function $vF(v)$ strictly decreases for $v > 0$.*

One concavity argument supplies the needed entropy bound. The function $(1 - r^2)g(r) - rh(r)$ has continuous value zero at both endpoints of $[0, 1]$, and its second derivative is $-r/(1 - r^2) < 0$. It is therefore positive inside the interval, giving

$$h(r) < \frac{(1 - r^2)g(r)}{r}.$$

Now fix $v > 0$ and put $r = \tanh v$, so $g(r) = v$. The factor $r - v(1 - r^2)$ is positive: it vanishes at zero and its derivative

in v is $2vr(1-r^2) > 0$. We may consequently use the entropy bound in the derivative

$$\begin{aligned} \frac{d}{dv}[vF(v)] &= \frac{h(r)[r-v(1-r^2)]}{r^2} - \frac{v^2(1-r^2)}{r} \\ &< \frac{v(1-r^2)}{r^3}(r-v) < 0, \end{aligned}$$

where the last step uses $\tanh v < v$.

a) Consequence for the small-height comparison.:

Choose an upper height $0 < v_{\text{cut}} \leq \ell_\rho$. For $0 < v \leq v_{\text{cut}}$ and $a = F(\ell_\rho)/F(v)$, monotonicity gives $a/v \leq F(\ell_\rho)/[v_{\text{cut}}F(v_{\text{cut}})]$. Also $c(v) \leq v$, since $\text{corr} \geq 0$. The LSI branch of Lemma 7 now gives

$$\text{yield}(v) \leq \rho + pK_{\text{LS}}\rho^2 \frac{F(\ell_\rho)}{v_{\text{cut}}F(v_{\text{cut}})} \quad (0 < v \leq v_{\text{cut}}). \quad (\text{S5})$$

This bounds the entire small- v interval by one expression.

C. Relative Rates of the Entropy Profile and Angle Correction

The middle argument compares the profile $vF(v)$ with the edge correction $v - c(v)$. The following statement involves only these two functions, independently of the channel or the Boolean rule.

Lemma 21 (The profile pays for the angle correction). *Define the logarithmic decay rate $Q(v) = -F'(v)/F(v)$. Then*

$$\begin{aligned} Q(v) - \frac{1}{v} &> \frac{3}{2}[1 - c'(v)] \quad (v > 0), \\ Q(v) - \frac{1}{v} &> 2[1 - c'(v)] \quad (0 < v \leq 2). \end{aligned} \quad (48)$$

Equivalently, the rate of decrease of $\log[vF(v)]$ exceeds three halves of the rate of increase of $v - c(v)$; up to height two, the factor improves to two.

Proof. We first record the angle geometry used in the rate comparison. For a variable height $v > 0$, write $r_v = \tanh v$, $\tau_v = \text{sech } v$, $\theta_v = \arcsin r_v$, and $s(v) = \theta_v/\sinh v$. Since $\theta'_v = \tau_v$, we have $0 < \theta_v < v < \sinh v$, hence $0 < s < 1$. Differentiation gives

$$s' = \frac{\tau_v^2 - s}{r_v} < 0, \quad c' = 2s - s^2, \quad c'' = 2(1-s)s' < 0.$$

The first sign follows from $\theta_v > r_v\tau_v$. Thus c increases, and $c(v)/v$ decreases because c is concave and $c(0) = 0$.

We will use the two angle bounds

$$\frac{3z}{2 + \sqrt{1-z^2}} \leq \arcsin z \leq \frac{\pi z}{2 + \sqrt{1-z^2}} \quad (0 \leq z \leq 1). \quad (49)$$

Here is a proof of both at once. Set $z = \sin t$, $0 < t < \pi/2$. The derivative of $t(2 + \cos t)/\sin t$ has numerator $(2 + \cos t)\sin t - t(1 + 2\cos t)$. This numerator vanishes at zero and has derivative $2\sin t(t - \sin t) > 0$. The quotient therefore increases from three to π , proving the bounds.

Since $1 - c' = (1 - s)^2 > 0$, it suffices to compare $Q - 1/v$ with the indicated multiples of $(1 - s)^2$. The fixed estimate $e^4 < 55$ below follows from the finite sums in Section F-G.

Two simple lower bounds on Q suffice. With $q_v = e^{-2v}$, the entropy formula gives

$$Q(v) = \frac{4q_v[v + \log(1 + q_v)]}{(1 - q_v)[(1 + q_v)\log(1 + q_v) + 2vq_v]} \geq \frac{4v}{2v + 1}.$$

Indeed $\log(1 + q_v) \leq q_v$ bounds the denominator by $q_v(1 + 2v)$, while the numerator is at least $4q_v v$. Also $Q(v) > \coth v$. To see this, let

$$\begin{aligned} \mathcal{H}(v) &= \cosh v h(\tanh v), \\ D(v) &= v \coth v - \log(2 \cosh v). \end{aligned}$$

We have $D' = (\tanh v - v)/\sinh^2 v < 0$ and $D(v) \rightarrow 0$ at infinity, so $\mathcal{H}' = -\sinh v D < 0$. Since $F = \mathcal{H}/\sinh v$, this proves the claimed bound.

The positive power series show that $(\cosh v - 1)/v^2$ and $\sinh v/v$ increase. The geometric estimates

$$\begin{aligned} \cosh 1 - 1 &< \frac{1/2}{1 - 1/12} < \frac{3}{5}, \\ \sinh 1 - 1 &< \frac{1/6}{1 - 1/20} < \frac{1}{5}, \\ \cosh 2 - 1 &< \frac{2}{1 - 1/3} = 3 \end{aligned}$$

and $\cosh 4 < (55 + 1)/2 < 33$ will give convenient quadratic bounds. For $0 < v \leq 1$, they imply $\cosh v \leq 1 + 3v^2/5$. By (49),

$$1 - s \leq \frac{2v^2}{5 + 2v^2}, \quad Q(v) - 1/v > \coth v - 1/v > \frac{5v}{18} > \frac{v}{4}.$$

The second inequality uses $v \cosh v - \sinh v = \int_0^v t \sinh t dt \geq v^3/3$ and $\sinh v < 6v/5$. Finally $(5 + 2v^2)^2 - 32v^3 \geq 25 - 12v^2 > 0$, so $v/4 > 2[2v^2/(5 + 2v^2)]^2$.

For $1 \leq v \leq 2$, use $\cosh v \leq 1 + 3v^2/4$, giving $1 - s \leq v^2/(2 + v^2)$. The numerator of

$$\frac{4v}{2v + 1} - \frac{1}{v} - 2\frac{v^4}{(2 + v^2)^2},$$

after multiplication by $v(2v + 1)(2 + v^2)^2$, is

$$v^3(15v - 4v^2 - 8) + 4(3v + 1)(v - 1) > 0.$$

The concave quadratic in the first term is at least three on $[1, 2]$; the second term is nonnegative. This proves the stronger rate bound for $0 < v \leq 2$.

For $2 \leq v \leq 4$, the bound $\cosh v \leq 1 + 2v^2$ gives $1 - s \leq 4v^2/(3 + 4v^2)$. The numerator of

$$\frac{4v}{2v + 1} - \frac{1}{v} - \frac{3}{2}\frac{16v^4}{(3 + 4v^2)^2},$$

after multiplication by $v(2v + 1)(3 + 4v^2)^2$, is

$$16v^4(v - 2)(v - 3/2) + 16v^3(2v - 3) + 3(4v^2 - 6v - 3) > 0.$$

Each term is nonnegative, and the last quadratic is increasing from one. For $v \geq 4$, simply use $Q(v) - 1/v \geq 2 - 2/(2v + 1) - 1/v > 3/2$ and $(1 - s)^2 < 1$. This proves the global rate bound. $\square$

To use this lemma, integrate its rate bound between a coordinate height and the channel height. Appendix C checks which coefficient the channel needs; the profile calculation itself is now complete.

D. Tensorized Logarithmic-Sobolev Inequality

The coordinate estimate in Lemma 7 uses

$$\text{Ent}(w^2) \leq 2\mathbb{E}[w\mathcal{L}w].$$

This classical inequality measures how far the nonnegative field w^2 is from constant by the squared differences of w across edges. We include its proof, including the step that makes its constant independent of dimension.

Start with one bit. Replacing the two values of w by their absolute values preserves entropy and reduces their squared difference. Homogeneity then lets us assume $\mathbb{E}w^2 = 1$ and write the values as $\sqrt{1+s}, \sqrt{1-s}$, where $0 \leq s \leq 1$. In this notation

$$\begin{aligned} \text{Ent}(w^2) &= \phi(s), \\ 2\mathbb{E}[w\mathcal{L}w] &= \frac{(w_+ - w_-)^2}{2} = 1 - \sqrt{1-s^2}. \end{aligned}$$

The derivative of $s/\sqrt{1-s^2} - \text{atanh } s$ is $(1-s^2)^{-3/2} - (1-s^2)^{-1} \geq 0$. The difference vanishes at zero. Integrating $\phi'(s) = \text{atanh } s$ therefore proves $\phi(s) \leq 1 - \sqrt{1-s^2}$. Continuity supplies $s = 1$.

It remains to tensorize the one-bit bounds. For a finite probability space, entropy is convex as a function of a nonnegative vector z . For positive z , its second derivative in a direction v is

$$\mathbb{E} \frac{v^2}{z} - \frac{(\mathbb{E}v)^2}{\mathbb{E}z} \geq 0$$

by Cauchy-Schwarz. Zero entries follow by continuity. For independent coordinates X, Y , the entropy chain rule and this convexity give

$$\begin{aligned} \text{Ent}_{X,Y}(z) &= \mathbb{E}_Y \text{Ent}_X(z) + \text{Ent}_Y(\mathbb{E}_X z) \\ &\leq \mathbb{E}_Y \text{Ent}_X(z) + \mathbb{E}_X \text{Ent}_Y(z). \end{aligned}$$

Induction over the coordinates yields $\text{Ent}(z) \leq \sum_i \mathbb{E} \text{Ent}_i(z)$, where Ent_i takes entropy over coordinate i with all others fixed. Apply the one-bit result on each such pair, with $z = w^2$:

$$\text{Ent}(w^2) \leq \frac{1}{2} \sum_i \mathbb{E}_{-i} (w_{i,+} - w_{i,-})^2 = 2\mathbb{E}[w\mathcal{L}w].$$

No constant was multiplied by the number of coordinates.

E. Propagation of the Local Cutoff

This is the threshold-propagation property in Yu's Lemma 4.9 [11], expressed in sign means and natural logarithms. We include the one-sign-change proof so that the local comparison requires no numerical monotonicity test or external lemma.

Define the scalar local margin

$$M_{\text{loc}}(r, a) = (1-r^2)h(ar) - (1-ar^2)h(r), \quad 0 \leq r, a \leq 1.$$

We prove the propagation used below: if $M_{\text{loc}}(r_1, a) \geq 0$ for some $0 < r_1 < 1$, then $M_{\text{loc}}(r, a) \geq 0$ whenever $0 \leq r \leq r_1$.

The reason is a single change of sign in its entropy-series coefficients.

The case $a = 1$ is an identity. For $a < 1$, put $t = r^2$ and $H(t) = h(\sqrt{t})$. Let $b_n = 1/[2n(2n-1)]$ be the entropy-series coefficient, and let $T_n = \sum_{j>n} b_j$ be its tail. Then

$$H(t) = L - \sum_{n \geq 1} b_n t^n,$$

$$\frac{H(t)}{1-t} = \sum_{n \geq 0} T_n t^n,$$

$$T_n = \int_0^1 \frac{x^{2n}}{1+x} dx.$$

For the last identity, express $b_j = \int_0^1 x^{2j-2}(1-x) dx$, sum the geometric series, and cancel $1-x$. All series used here converge absolutely for $t < 1$.

Expansion of the local margin now gives

$$\begin{aligned} \frac{M_{\text{loc}}(\sqrt{t}, a)}{1-t} &= \sum_{n \geq 1} q_n t^n, \\ q_n &= \frac{1-a}{2n} \left[\frac{a + a^2 + \dots + a^{2n-1}}{2n-1} - 2nT_n \right]. \end{aligned}$$

The average of the powers of a decreases with n : the next average appends two terms no larger than the preceding terms. In the opposite direction,

$$(n+1)T_{n+1} - nT_n = T_n - \frac{1}{2(2n+1)} > 0,$$

as is seen by replacing $1/(1+x)$ by $1/2$ in the integral. Thus the coefficients q_n are nonnegative first and negative thereafter, with at most one change of sign. They eventually are negative, since the power average tends to zero whereas $2nT_n > n/(2n+1)$.

Let N be the last index with $q_N \geq 0$. Such an index exists if the margin is nonnegative at $r_1 > 0$. For $t_1 = r_1^2$ and $\lambda = t/t_1 \in (0, 1]$, the sign ordering gives

$$\sum_{n \geq 1} q_n t^n \geq \lambda^N \sum_{n \geq 1} q_n t_1^n \geq 0.$$

Indeed, $\lambda^n \geq \lambda^N$ before the sign change; afterwards the inequality reverses when multiplied by the negative coefficient. This proves the propagation, with $r = 0$ immediate.

Finally $\partial_a^2 M_{\text{loc}}(r, a) = -(1-r^2)r^2/(1-a^2r^2) < 0$ and $M_{\text{loc}}(r, 1) = 0$. A successful check at (r_1, a_0) therefore covers all $r \leq r_1$ and $a \geq a_0$: concavity extends in a , and the series argument extends in r .

F. The Influential-Coordinate Criterion

The starting point is the standard strong data-processing bound obtained from cube logarithmic Sobolev [29], [31]. The local-optimality perspective is shared with [6], [11]; the calculation below makes explicit the mean correction needed to use this particular criterion for biased rules.

For an arbitrary Boolean f , fix a coordinate and write $a = |\hat{f}(\{i\})|$, $m = Ef$, $\delta = 1 - \rho^2$. Necessarily $|m| + a \leq 1$.

For a cube field v taking values in $[-1, 1]$, define $\text{Ent}_\phi(v) = \mathbb{E}\phi(v) - \phi(\mathbb{E}v)$. We first derive the entropy contraction used here and in Appendix D:

$$\text{Ent}_\phi(T_\rho v) \leq \rho^2 \text{Ent}_\phi(v). \quad (\text{S6a})$$

For a positive field w , the elementary edge inequality

$$(p - q)(\log p - \log q) \geq 4(\sqrt{p} - \sqrt{q})^2 \quad (p, q > 0)$$

follows by Cauchy–Schwarz applied to the integrals of 1 and $1/\sqrt{t}$ between p and q . Sum over cube edges and use the logarithmic Sobolev inequality just proved in Section A-D:

$$\mathbb{E}[(\mathcal{L}w) \log w] \geq 4\mathbb{E}[\sqrt{w} \mathcal{L}\sqrt{w}] \geq 2 \text{Ent}(w).$$

Let $t \geq 0$ be the noise-flow time and define $w_t = T_{e^{-t}}w$. The mean is constant and $\frac{d}{dt} \text{Ent}(w_t) = -\mathbb{E}[(\mathcal{L}w_t) \log w_t]$. Integration gives $\text{Ent}(T_\rho w) \leq \rho^2 \text{Ent}(w)$. For a positive real ε , nonnegative fields follow by replacing w with $w + \varepsilon$ and taking $\varepsilon \downarrow 0$. Finally,

$$\text{Ent}_\phi(v) = \frac{1}{2} \{ \text{Ent}(1 + v) + \text{Ent}(1 - v) \},$$

which proves (S6a).

Apply (S6a) to the other coordinates after noising the chosen bit first. The intermediate field has values $\pm\rho$ on pivotal edges and ± 1 otherwise. Pivotal probability is at least a , and the section means are $m \pm \rho a$. Thus

$$Eh(T_\rho f) \geq \rho^2 a h(\rho) + \frac{\delta}{2} [h(m + \rho a) + h(m - \rho a)]. \quad (\text{S6})$$

The mean costs no additional loss because

$$\phi(m) + \frac{\delta}{2} [h(m + \rho a) + h(m - \rho a)] \geq \delta h(\rho a). \quad (\text{S7})$$

For completeness, twice differentiate the difference in m . After positive factors are removed, its second derivative has the sign of

$$[1 - (m + a)^2][1 - (m - a)^2] + (1 - \rho^2)a^2(1 - m^2 - a^2) \geq 0.$$

The difference and its first derivative vanish at zero. Integrating proves (S7); reflection handles negative m .

Combining (S6)–(S7) proves CK whenever

$$(1 - \rho^2)h(\rho a) - (1 - \rho^2 a)h(\rho) \geq 0. \quad (\text{S8})$$

For fixed ρ , the left side is concave in a and zero at $a = 1$. Checking it at a cutoff therefore covers every larger a . Section A-E proves directly why a successful cutoff also works at smaller correlations. The all-bias step is (S7), proved here.

APPENDIX B

AN ENTROPY TANGENT FOR THE ANGULAR ERROR

This appendix proves (10). Its purpose is to preserve equality at the dictator's two posterior values $\pm r$, while controlling the error of the line $x \mapsto (\arcsin r)x/r$.

Fix $5/4 \leq \lambda \leq \pi/2$. On $0 \leq x \leq 1$, define

$$\begin{aligned} A_{\text{sc}}(x) &= x \arcsin x, \\ e(x) &= \arcsin x - \lambda x, \\ \Psi(x) &= \phi(x) + e(x)^2/2. \end{aligned}$$

We will show that Ψ increases and that A_{sc} is convex as a function of Ψ . A tangent at $x = r$ will then give exactly the required inequality.

Put $x = \sin t$, $c = \cos t$, and $g = \text{atanh}(\sin t)$. Dots in this paragraph denote derivatives in t . Direct differentiation gives

$$\begin{aligned} \dot{A}_{\text{sc}} &= ct + x, & \ddot{A}_{\text{sc}} &= 2c - xt, \\ \dot{\Psi} &= gc + e(1 - \lambda c), \\ \ddot{\Psi} &= 1 - gx + (1 - \lambda c)^2 + \lambda x e. \end{aligned}$$

The terms linear in λ cancel in the curvature numerator:

$$K_\lambda := \ddot{A}_{\text{sc}} \dot{\Psi} - \dot{A}_{\text{sc}} \ddot{\Psi} = g(1 + c^2) - x(t^2 + 2) + \lambda^2(x - c^3 t).$$

Since $t < \tan t = x/c$, its last coefficient is positive. It suffices to show $K_{5/4} > 0$.

Here are two matched integral bounds, with their remainders exposed:

$$\begin{aligned} \left[\frac{y(15 + 4y^2)}{15 + 9y^2} - \arctan y \right]' &= \frac{36y^6}{(15 + 9y^2)^2(1 + y^2)}, \\ \left[\text{atanh } y - \frac{y(15 - 4y^2)}{15 - 9y^2} \right]' &= \frac{36y^6}{(15 - 9y^2)^2(1 - y^2)}. \end{aligned}$$

Both differences vanish at zero. Substituting $y = \tan(t/2)$ and using the half-angle formulas gives

$$\frac{t}{x} \leq \frac{19 + 11c}{3(1 + c)(4 + c)}, \quad \frac{g}{x} \geq \frac{11 + 19c}{3(1 + c)(1 + 4c)}.$$

Use these upper bounds for the negative t, t^2 terms and the lower bound for g . After multiplication by positive denominators the result is

$$\frac{16K_{5/4}}{x} \geq \frac{(1 - c)Q(c)}{9(1 + c)(4 + c)^2(1 + 4c)},$$

where the following identity proves the sign directly:

$$\begin{aligned} Q(c) &= 375(2 - 6c - 7c^2)^2 + 164(1 - c)^5 \\ &\quad + 740c(1 - c)^4 + c^2(1 - c)^3 \\ &\quad + 25c^3(1 - c)^2 + 5925c^4(1 - c) + 8625c^5. \end{aligned}$$

Indeed expansion gives $Q(c) = 1664 - 9080c + 1681c^2 + 34322c^3 + 22113c^4 + 3300c^5$, the numerator produced by the preceding substitution. Each term in the identity is nonnegative on $0 \leq c \leq 1$, and not all vanish. Thus $K_\lambda > 0$ in the interior.

To justify using Ψ as a coordinate, observe that

$$\left(\frac{\dot{\Psi}}{\dot{A}_{\text{sc}}} \right)' = -\frac{K_\lambda}{\dot{A}_{\text{sc}}^2} < 0.$$

As $t \uparrow \pi/2$, this ratio tends to $\pi/2 - \lambda \geq 0$. It is therefore positive before the endpoint. This proves $\dot{\Psi} > 0$, and $d^2 A_{\text{sc}}/d\Psi^2 = K_\lambda/\dot{\Psi}^3 > 0$.

Now let $r = \tanh \ell$, $\ell \geq 31/20$, and $\lambda = \arcsin(r)/r$. This slope lies in $[5/4, \pi/2]$. For the lower bound, $e^{31/10} > 20(1 + 1/10) = 22$ gives $r > 21/23$, hence $r^2 > 5/6$. The alternating sine bound, with $y = 5r/4 < 5/4$, gives

$$\frac{\sin(5r/4)}{r} \leq \frac{5}{4} - \frac{125}{384}r^2 + \frac{3125}{122880}r^4 < 1.$$

The quadratic on the right decreases on $0 \leq r^2 \leq 1$; its value at $5/6$ is below one. As $5r/4 < \pi/2$, $\arcsin r > 5r/4$. The upper bound is the endpoint chord of the convex function $\arcsin r$.

At $x = r$ we have $e(r) = 0$, $\Psi(r) = \phi(r)$, and

$$\frac{dA_{sc}}{d\Psi}(r) = \frac{\arcsin r + r/\sqrt{1-r^2}}{\operatorname{atanh} r} = \alpha.$$

The supporting line of the proved convex function is $A_{sc}(x) - A_{sc}(r) \geq \alpha[\phi(x) - \phi(r) + e(x)^2/2]$. This is (10); evenness extends it to negative x , and continuity includes the endpoints.

APPENDIX C

CHANNEL-PARAMETER BOUNDS IN THE INTERMEDIATE REGIME

Lemma 22 (The profile, energy, and mean reserves). *Let $31/20 \leq \ell \leq 7/2$, and put*

$$\begin{aligned} r &= \tanh \ell, & \theta &= \arcsin r, & \lambda &= \frac{\theta}{r}, \\ \alpha &= \frac{\theta + \sinh \ell}{\ell}, & \gamma &= \lambda \alpha, \\ \eta &= \frac{\theta^2 \gamma}{\gamma + 1}, & \beta &= \frac{2r}{\eta}. \end{aligned}$$

Furthermore, define

$$\begin{aligned} F(v) &= \frac{h(\tanh v)}{\tanh v}, \\ c(v) &= \frac{\arcsin(\tanh v)^2}{\tanh v}, \\ H(v) &= v - c(v). \end{aligned}$$

Then the following three inequalities hold:

$$\log \frac{vF(v)}{\ell F(\ell)} \geq \beta[H(\ell) - H(v)] \quad (0 < v \leq \ell), \quad (50)$$

$$\theta^2 - rH(\ell) - \frac{\eta}{2} \geq 0, \quad (51)$$

$$\gamma - \frac{2\theta^2}{\sinh^2 \ell} - \frac{\lambda}{\alpha} \geq r\ell. \quad (52)$$

Proof. The profile geometry is isolated in Lemma 21. In particular, c increases, $c(v)/v$ decreases, and the two ratios

$$s(v) = \frac{\arcsin(\tanh v)}{\sinh v}, \quad \lambda(v) = \frac{\arcsin(\tanh v)}{\tanh v}$$

are respectively decreasing and increasing. The latter assertion follows because the derivative of $\arcsin$ is increasing, so $\arcsin r < r/\sqrt{1-r^2}$. Also $\alpha(v)$ is the average on $[0, v]$ of $\operatorname{sech} t + \cosh t$, whose derivative is $\sinh t \tanh^2 t > 0$. Thus $\alpha > 2$, and both α and γ increase.

The two angle bounds (49) were proved with the global profile lemma. We also use

$$z - \frac{z^3}{3} \leq \arctan z \leq z - \frac{z^3}{3(1+z^2)} \quad (z \geq 0). \quad (53)$$

These follow by bounding the integrand in $z - \arctan z = \int_0^z t^2/(1+t^2) dt$.

All fixed constant bounds below follow from the finite sums and geometric remainders in Section F-G. In particular we use

$$\begin{aligned} \frac{157}{50} < \pi < \frac{22}{7}, & \quad 20 < e^3 < \frac{81}{4}, \\ e^4 < 55, & \quad e^{31/20} > \frac{80}{17}, & \quad e^{1/2} > \frac{8}{5}. \end{aligned}$$

a) *The profile check: two channel endpoints.*: Lemma 21 supplies the rate bounds at every height. We only need to bound their channel coefficient. The identity

$$\beta(\ell) = \frac{2}{c(\ell)} \left(1 + \frac{1}{\gamma(\ell)} \right)$$

shows that β decreases. Its two needed endpoint bounds are $\beta(31/20) < 2$ and $\beta(2) < 3/2$, as follows. Monotonicity of λ , the angle identity $\theta = \pi/2 - 2 \arctan(e^{-\ell})$, and $e^{-31/20} < 17/80$ give

$$\lambda(31/20) > \left(\frac{157}{100} - \frac{17}{40} \right) \frac{1 + (17/80)^2}{1 - (17/80)^2} > \frac{5}{4}.$$

Also $r > 19/21$, by $e^3 > 20$, and $\alpha > 2$. Thus $c(31/20) > 475/336 > 7/5$, $\gamma(31/20) > 5/2$, and $\beta(31/20) < 2$.

At two, the elementary series bounds $155/21 < e^2 < 37/5$ give $r < 27/28$ and $\sinh 2 > 18/5$. The lower bound is the exponential sum through degree seven; the upper bound follows from the sum through degree four and its geometric remainder. With $t = 21/155$, (53) gives

$$\arctan(e^{-2}) < t - \frac{t^3}{3(1+t^2)} < \frac{27}{200}, \quad \theta(2) > \frac{13}{10}.$$

These imply $c(2) > 7/4$, $\lambda(2) > 4/3$, $\alpha(2) > 49/20$, and $\gamma(2) > 13/4$. Consequently $\beta(2) < 136/91 < 3/2$. Combining these endpoint bounds with Lemma 21 proves $Q(v) - 1/v \geq \beta(\ell)[1 - c'(v)]$ for every $v \leq \ell$. Integration from v to ℓ is exactly (50).

b) *The energy check.*: Since $rH(\ell) = r\ell - \theta^2$, condition (51) is equivalent to

$$\frac{c(\ell)}{\ell} \left[\frac{3}{2} + \frac{1}{2(\gamma+1)} \right] \geq 1.$$

Both positive factors decrease, so only $\ell = 7/2$ needs checking. The fixed exponential bounds give $32 < e^{7/2} < 35$. At that height,

$$\begin{aligned} \theta &> \frac{\pi}{2} - \frac{1}{16}, & \lambda &< \frac{8}{5}, \\ \alpha &< \frac{8/5 + 35/2}{7/2} = \frac{191}{35}, & \gamma &< 9. \end{aligned}$$

The angle bound uses $\arctan z < z$; the bound on λ also follows from (49). Since $c = \theta^2/r > \theta^2$, the required product exceeds

$$\frac{31}{20} \frac{(\pi/2 - 1/16)^2}{7/2} > 1.$$

The last rational comparison already follows from $\pi > 157/50$.

c) *The mean check: a positive completed square.*: We prove (52) on the larger range $\ell \geq 3/2$. Put $\tau = \text{sech } \ell$, and continue to use $s = s(\ell)$. At $\ell_0 = 3/2$, the fixed exponential bounds and the positive cosh series give

$$\frac{19}{21} < r < \frac{77}{85}, \quad \frac{75}{32} < \cosh \ell_0 < \frac{5}{2}, \quad (54)$$

$$\frac{19}{9} < \sinh \ell_0 < \frac{77}{36}.$$

For example, the lower sinh bound is $\sinh \ell_0 > 19/(4\sqrt{5}) > 19/9$, and the lower cosh bound uses its terms through degree six. The upper cosh bound follows from $e^{\ell_0} < 9/2$ and $e^{-\ell_0} < 1/4$. The two angle bounds at this point are

$$\frac{9}{8} < \theta(\ell_0) < \frac{8}{7}.$$

Indeed, $e^{-\ell_0} < 1/(2\sqrt{5})$ and (53) give $\theta(\ell_0) > \pi/2 - 62/(63\sqrt{5}) > 9/8$; the last comparison follows from $\pi > 157/50$ by squaring positive quantities. Conversely $e^{-\ell_0} > 2/9$ gives $\theta(\ell_0) < 11/7 - 4/9 + 16/2187 < 8/7$. In particular

$$\frac{81}{154} < s(\ell_0) < \frac{72}{133},$$

$$2s(\ell)^2 < \frac{3}{5} \quad (\ell \geq \ell_0),$$

where the uniform bound follows from the decrease of s .

To bound the positive terms in (52), define

$$Y(\ell) = c(\ell) + \theta \cosh \ell - r\ell^2 - \frac{6}{5}\ell.$$

Thus $Y/\ell = \gamma - r\ell - 6/5$. Differentiating the angle identities above gives

$$Y' = 2s - s^2 + \theta \sinh \ell - \tau^2 \ell^2 - 2r\ell - \frac{1}{5}.$$

With $T = \theta \cosh \ell - r$, another derivative gives the exact factorization

$$Y'' = T \left[1 - \frac{2(1-s)}{\sinh^2 \ell} \right] + 2\tau^2 \ell(r\ell - 2).$$

Throughout $\ell \geq 3/2$, $r > 9/10$, $\cosh \ell > 7/3$, and $\sinh^2 \ell > 40/9$. The lower angle bound gives $\lambda > 6/5$, since $\tau < 1/2$. Hence $T > 81/50$, and its bracket exceeds $11/20$. Also $\tau^2 \ell$ decreases, because its derivative is $\tau^2(1-2r\ell) < 0$. Its initial value is less than $(9/49)(3/2) < 3/10$. As $r\ell > 27/20$, the second term exceeds $-39/100$, whether it is negative or nonnegative. Therefore

$$Y'' > \frac{81}{50} \frac{11}{20} - \frac{39}{100} > \frac{1}{2}.$$

The endpoint bounds already established give

$$Y(\ell_0) > \left(\frac{9}{8}\right)^2 \frac{85}{77} + \frac{9}{8} \frac{75}{32} - \frac{77}{85} \frac{9}{4} - \frac{9}{5} > \frac{3}{16},$$

$$Y'(\ell_0) > 2 \frac{81}{154} - \left(\frac{81}{154}\right)^2 + \frac{19}{8} - \frac{20}{49} - \frac{231}{85} - \frac{1}{5} > -\frac{1}{5}.$$

For the second line, use that $2s - s^2$ increases for $0 < s < 1$, and $\tau(\ell_0)^2 < 80/441$. Integrating the curvature bound, with $t = \ell - 3/2 \geq 0$, gives

$$Y(\ell) > \frac{3}{16} - \frac{t}{5} + \frac{t^2}{4} = \frac{1}{4} \left(t - \frac{2}{5} \right)^2 + \frac{3}{16} - \frac{1}{25} > 0.$$

Thus $\gamma - r\ell > 6/5$.

It remains to bound λ/α . The upper angle bound in (49) implies

$$\frac{\lambda}{\alpha} = \frac{\ell\theta}{r(\theta + \sinh \ell)} \leq \frac{\pi\ell}{(\pi+1)r + 2\sinh \ell}.$$

Define $Z(\ell) = (\pi+1)r + 2\sinh \ell - (5\pi/3)\ell$. For $\ell \geq 3/2$,

$$Z'' = 2\sinh \ell - 2(\pi+1)r\tau^2 > 4 - \frac{522}{343} > 0.$$

Here $\tau^2 < 9/49$ and $\pi+1 < 29/7$. At the initial point, (54) gives

$$Z'(\ell_0) > \frac{16}{25} + \frac{14}{3} - \frac{110}{21} > 0,$$

$$Z(\ell_0) > \frac{19}{21} + 2\frac{19}{9} - \frac{67}{42} \frac{22}{7} > 0.$$

The derivative bound uses $\tau(\ell_0)^2 > 4/25$, and the value bound combines the terms containing π before using its upper bound. Consequently $Z > 0$, proving $\lambda/\alpha < 3/5$. Together with the bounds above,

$$\gamma - r\ell - 2s^2 - \frac{\lambda}{\alpha} > \frac{6}{5} - \frac{3}{5} - \frac{3}{5} = 0.$$

This proves (52), and completes all three checks. $\square$

APPENDIX D THE SMALL-CORRELATION REGIME

This appendix proves CK for $0 \leq \rho \leq \rho_*$, at every bias, where $\rho_* = \tanh(31/20)$ is the common boundary. Its mechanism is an energy gap. A rule with one large singleton coefficient is covered by the local lemma. Otherwise its total singleton energy is bounded away from one. Two quadratics tangent to binary entropy convert that energy gap into the required information bound. This is the local-optimality/Fourier-gap strategy developed in Samorodnitsky's and Yu's work [6], [10], [11]. The distinction in this appendix is its self-contained quantitative input: the cap below is proved by sections and sign sums, and the entropy comparison keeps the output mean. Yu's published balanced range is not being used as an all-bias black box.

Write $r = \rho$ in this appendix. For a cube function v , its *singleton energy* is $W_1(v) = \sum_i \hat{v}(\{i\})^2$. Let $a = \max_i |\hat{f}(\{i\})|$ be the largest singleton coefficient in absolute value. Use the cutoff $a_0 = 5/8$ and the energy cap

$$M = 1 - a_0 + a_0^2 = \left(\frac{7}{8}\right)^2.$$

These choices match two bounds: splitting on the strongest coordinate gives $1 - a_0 + a_0^2$, while the sign-sum estimate below gives $7/8$. The cutoff makes their energy bounds agree exactly. We shall start the entropy comparison at $r_0 = 5/8$ and end at $r_1 = \rho_*$. Contraction will then cover all correlations below r_0 .

A. An elementary bound on singleton energy

We prove the Fourier estimate needed below, including the estimate for sums of independent signs on which it rests. Let f be any function on the cube with values in $\{-1, 1\}$, and put

$$a_i = \hat{f}(\{i\}), \quad a = \max_i |a_i|, \quad W = \sum_i a_i^2 = W_1(f).$$

Thus a measures the strongest correlation with a single input bit, and W is the total squared correlation with all individual bits. We will show

$$a \leq \frac{5}{8} \implies W \leq 1 - a_0 + a_0^2 = M. \quad (55)$$

The estimate holds at every mean. It separates into two simple cases: one coefficient is moderately large, or every coefficient is small. This is a quantitative instance of the level-one rigidity underlying FKN [5], tailored to a fixed cutoff rather than an asymptotic closeness theorem. For the broader sign-sum problem see [32], and for a sharp small-coefficient asymptotic see [12]. The particular constants needed here are established below.

1) *A moderately large coefficient leaves little other energy:* Orient a chosen input bit so that its coefficient is $a \geq 0$, and let f_+, f_- be the two sections obtained by fixing that bit. On the remaining cube define

$$s = \frac{f_+ + f_-}{2}, \quad d = \frac{f_+ - f_-}{2}.$$

Pointwise, $s^2 + d^2 = 1$ and $d \in \{-1, 0, 1\}$. Also $\mathbb{E}d = a$, so $\mathbb{E}d^2 \geq a$. The other singleton coefficients are the singleton coefficients of s . Parseval therefore gives

$$W = a^2 + W_1(s) \leq a^2 + \mathbb{E}s^2 \leq a^2 + 1 - a. \quad (56)$$

For $3/8 \leq a \leq 5/8$, the convex quadratic on the right is at most its equal endpoint values:

$$\left(\frac{3}{8}\right)^2 + 1 - \frac{3}{8} = \left(\frac{5}{8}\right)^2 + 1 - \frac{5}{8} = \left(\frac{7}{8}\right)^2 = M.$$

It remains to handle $a \leq 3/8$.

2) *Spread-out coefficients cannot have absolute mean near one:* We use the following elementary moment bound. If $Z = \sum_i b_i X_i$, where the X_i are independent uniform signs, then

$$\sum_i b_i^2 = 1, \quad \max_i b_i^2 \leq \frac{1}{5} \implies \mathbb{E}|Z| < \frac{7}{8}. \quad (57)$$

The normalization gives $\mathbb{E}Z^2 = 1$, so Cauchy–Schwarz alone would give only $\mathbb{E}|Z| \leq 1$. The condition on the largest summand gives a strict gap. We prove it with one polynomial above absolute value; no Gaussian approximation or tail estimate is needed.

a) *Contact polynomial.:* Let p be the polynomial of degree at most five that matches $\sqrt{y}$ and its derivative at the three points

$$y = 9/16, \quad 4, \quad 9.$$

These are convenient contacts for one fixed bound, not parameters of the channel. The contact conditions determine p uniquely: the difference of two such polynomials would have

six zeros, counting multiplicity, despite having degree at most five. For complete arithmetic, put $D_{\text{mom}} = 4042912500$. The polynomial is

$$\begin{aligned} D_{\text{mom}}p(y) = & 637696y^5 - 17928800y^4 \\ & + 191446315y^3 - 1008169920y^2 \\ & + 3660186285y + 1260006516. \end{aligned} \quad (58)$$

Substitution verifies the six contact conditions.

To prove that the contact polynomial is a global majorant, fix $y > 0$ away from the contact points. Subtract from $\sqrt{s} - p(s)$ the multiple of $(s - 9/16)^2(s - 4)^2(s - 9)^2$ that makes it zero at $s = y$. There are seven zeros counting multiplicity. Applying Rolle's theorem six times gives, for some $\xi > 0$,

$$\sqrt{y} - p(y) = \frac{(\sqrt{\cdot})^{(6)}(\xi)}{6!}(y - 9/16)^2(y - 4)^2(y - 9)^2 \leq 0,$$

since $(\sqrt{\cdot})^{(6)}(\xi) = -945/(64\xi^{11/2}) < 0$. The contact points and zero follow by continuity. Thus

$$|Z| \leq p(Z^2).$$

This sign argument is the reason for using a contact polynomial.

b) *Moment evaluation.:* Write $\sigma_j = \sum_i b_i^j$ for $j = 4, 6, 8, 10$. Expansion of the sum, keeping only even multiplicities, gives

$$\begin{aligned} \mathbb{E}Z^2 &= 1, & \mathbb{E}Z^4 &= 3 - 2\sigma_4, \\ \mathbb{E}Z^6 &= 15 - 30\sigma_4 + 16\sigma_6, \\ \mathbb{E}Z^8 &= 105 - 420\sigma_4 + 140\sigma_4^2 \\ &\quad + 448\sigma_6 - 272\sigma_8, \\ \mathbb{E}Z^{10} &= 945 - 6300\sigma_4 + 6300\sigma_4^2 \\ &\quad + 10080\sigma_6 - 6720\sigma_4\sigma_6 \\ &\quad - 12240\sigma_8 + 7936\sigma_{10}. \end{aligned}$$

One systematic way to obtain these identities is to differentiate $\mathbb{E}e^{tZ} = \prod_i \cosh(b_i t)$ at zero. The coefficients needed are

$$\begin{aligned} \log \cosh t &= \frac{t^2}{2} - \frac{t^4}{12} + \frac{t^6}{45} \\ &\quad - \frac{17t^8}{2520} + \frac{31t^{10}}{14175} + O(t^{12}); \end{aligned}$$

they follow by integrating the Taylor series of $\tanh t$.

Put $t = \sigma_4$ in the rest of this paragraph. The largest-weight condition implies

$$\begin{aligned} 0 \leq t \leq \frac{1}{5}, & & \sigma_6 \leq \frac{t}{5}, \\ \sigma_{10} \leq \frac{\sigma_8}{5}, & & \sigma_8 \geq t^3. \end{aligned}$$

For the last inequality, the numbers b_i^2 are probability weights: Jensen for the convex map $x \mapsto x^3$ gives $\sum_i b_i^2(b_i^2)^3 \geq (\sum_i b_i^2 b_i^2)^3$. Substitute the moment identities into (58), then apply these four inequalities. The result is the single cubic bound

$$D_{\text{mom}}\mathbb{E}p(Z^2) \leq T_{\text{mom}}(t),$$

where

$$\begin{aligned} T_{\text{mom}}(t) = & 3487476486 + 77364454t \\ & + 650389376t^2 - \frac{9583071744}{5}t^3. \end{aligned}$$

To check the substitution directions, the coefficient of σ_6 before using $\sigma_6 \leq t/5$ is $1459014320 - 4285317120t > 0$ on $[0, 1/5]$. After $\sigma_{10} \leq \sigma_8/5$, the coefficient of σ_8 is $-9583071744/5 < 0$, so Jensen is used in the correct direction.

Only one value of this cubic needs checking. Its derivative is concave, and is positive at both endpoints of $[0, 1/5]$:

$$\begin{aligned} T'_{\text{mom}}(0) &= 77364454 > 0, \\ T'_{\text{mom}}(1/5) &= \frac{13440810318}{125} > 0. \end{aligned}$$

Hence $T_{\text{mom}}(t) \leq T_{\text{mom}}(1/5)$. The remaining exact comparison is

$$7D_{\text{mom}} - 8T_{\text{mom}}(1/5) = \frac{119582002252}{625} > 0.$$

It follows that $\mathbb{E}|Z| \leq \mathbb{E}p(Z^2) < 7/8$, proving (57).

3) *Return to the Boolean function:* Suppose $a \leq 3/8$ and $W \geq M = (7/8)^2$, and normalize $b_i = a_i/\sqrt{W}$. Then

$$\sum_i b_i^2 = 1, \quad \max_i b_i^2 \leq \frac{(3/8)^2}{(7/8)^2} = \frac{9}{49} < \frac{1}{5}.$$

Booleanity supplies the bridge to the moment bound:

$$\sqrt{W} = \mathbb{E} \left[f \sum_i b_i X_i \right] \leq \mathbb{E} \left[\sum_i b_i X_i \right] < \frac{7}{8}.$$

This contradicts $\sqrt{W} \geq \sqrt{M} = 7/8$ directly. Thus small coefficients give $W < M$. Together with the section bound on $[3/8, 5/8]$, this proves (55).

B. The same cutoff covers larger coefficients

Recall the local margin M_{loc} from Section A-E. At our common boundary $r_1 = \tanh(31/20)$, the fixed comparison in Section D-H gives

$$M_{\text{loc}}(r_1, 5/8) = (1 - r_1^2)h(5r_1/8) - (1 - 5r_1^2/8)h(r_1) > 0.$$

The propagation argument there extends this one check to every $r \leq r_1$ and $a \geq 5/8$. Thus every rule is covered: a coefficient at least $5/8$ invokes the local lemma; otherwise the singleton energy is at most $(7/8)^2$.

C. Mean-Sensitive Energy Bound

The Fourier estimate just proved says

$$a \leq a_0 \implies W_1(f) \leq M. \quad (59)$$

The local check in Section D-B, followed by Section A-E, handles $a \geq a_0$ throughout $r \leq r_1$.

For a biased rule, introduce one new sign bit and form

$$G(1, x) = f(x), \quad G(-1, x) = -f(-x).$$

The new rule is balanced. Its singleton coefficient on the new bit is $m = \mathbb{E}f$; on the old bits it has the original singleton coefficients. Consequently, if $|m| \leq a_0$ and $a \leq a_0$,

$$m^2 + W_1(f) = W_1(G) \leq M. \quad (60)$$

Only this Fourier relation is needed from G . Larger biases are covered directly by entropy contraction:

$$I_f(r) \leq r^2 h(m) < r^2/2 \leq \phi(r),$$

Indeed, the first two terms of the entropy series give, when $|m| \geq a_0 = 5/8$,

$$\begin{aligned} h(m) &\leq h(a_0) \leq L - a_0^2/2 - a_0^4/12 \\ &< 7/10 - a_0^2/2 - a_0^4/12 < 1/2. \end{aligned}$$

The point $r = 0$ is immediate.

D. Tangent Quadratic Minorants

Remark 23 (Role of the quadratic minorant). The first two moments of the noisy rule have simple Fourier formulas. A quadratic below entropy lets us use those formulas directly. We construct the quadratic by making it touch entropy, rather than by fitting its coefficients to a numerical grid.

For a contact point $0 < t < 1$ and an entropy argument $0 \leq s \leq 1$, define

$$\begin{aligned} A_t &= \frac{\log(2/(1+t))}{(1-t)^2}, \\ z_t &= 2t - \frac{g(t)}{A_t}, \\ Q_t(s) &= A_t(1-s)(1+s-z_t). \end{aligned}$$

The coefficient A_t sets the scale; z_t sets the shape. They are determined by contact at t and vanishing at one. The identity $h(t) - (1-t)g(t) = \log(2/(1+t))$ gives

$$Q_t(t) = h(t), \quad Q'_t(t) = h'(t), \quad Q_t(1) = h(1) = 0.$$

Moreover, every such quadratic is a global lower bound:

$$Q_t(s) \leq h(s) \quad (0 \leq s \leq 1). \quad (61)$$

To see this, put $D_t = h - Q_t$. Its second derivative $D''_t(s) = 2A_t - 1/(1-s^2)$ strictly decreases. It must be positive at t ; otherwise D_t would decrease strictly from its zero at t to its zero at one. Thus D_t decreases to zero before t , then rises and falls back to zero on $[t, 1]$. It stays nonnegative throughout. This proves the entire family at once.

Use the two contact points $t_0 = 2/5$ and $t_1 = 2/3$. For $j = 0, 1$, abbreviate $A_j = A_{t_j}$ and $z_j = z_{t_j}$. Define α, β by the line through the two points $(z_j, 1/A_j)$:

$$\begin{aligned} \beta &= \frac{1/A_0 - 1/A_1}{z_1 - z_0}, \\ \alpha &= \frac{1}{A_0} + \beta z_0, \\ c_{\text{ent}}(r) &= \frac{1}{\alpha - \beta r^2}. \end{aligned}$$

Thus the coefficient is determined by the two entropy tangencies. The fixed comparisons at the end of this appendix give $z_0 < r_0^2$ and $r_1^2 < z_1$. For any $r \in [r_0, r_1]$, write $r^2 = \mu z_0 + (1-\mu)z_1$, where $0 \leq \mu \leq 1$. The weights $\mu c_{\text{ent}}/A_0$ and $(1-\mu)c_{\text{ent}}/A_1$ sum to one. Their convex combination of (61) gives

$$h(s) \geq c_{\text{ent}}(r)(1-s)(1+s-r^2). \quad (62)$$

This is the only entropy minorant used below.

E. From Entropy to Fourier Energy

For each Fourier degree k , put $W_k = \sum_{|S|=k} \hat{f}(S)^2$. Since f takes only the values $-1, 1$,

$$\mathbb{E}|T_r f| \geq \mathbb{E}[f T_r f] = \sum_{k=0}^n r^k W_k.$$

Apply (62) to $|T_r f|$, and use Parseval for its second moment. All higher-degree contributions have the favorable sign: $r^{k+2} - r^{2k} \geq 0$ for $k \geq 2$. Hence

$$\mathbb{E}h(T_r f) \geq c_{\text{ent}}(r)[1 - r^2 - (1 - r^2)m^2 - r^2(1 - r)W_1].$$

Insert (60) and $\phi(m) \geq m^2/2$:

$$\begin{aligned} \mathbb{E}h(T_r f) + \phi(m) &\geq c_{\text{ent}}(r)(1 - r)(1 + r - Mr^2) \\ &\quad + [1/2 - c_{\text{ent}}(r)(1 - 2r^2 + r^3)]m^2. \end{aligned} \quad (63)$$

The mean coefficient is positive. After multiplication by $2/c_{\text{ent}}(r) > 0$, it becomes

$$\alpha - 2 + (4 - \beta)r^2 - 2r^3.$$

This increases on $[r_0, 1]$, since $\beta < 1$. At $r_0 = 5/8$, the bounds $\alpha > 53/40$ and $\beta < 17/20$ give a value larger than $43/640 > 0$. We may therefore discard the mean term.

F. One change of curvature controls the remaining gap

The information inequality now reduces to the positivity of

$$B_0(r) = (1 - r)(1 + r - Mr^2) - (\alpha - \beta r^2)h(r).$$

We prove that B_0 decreases, so a single comparison at r_1 finishes the interval. The reason is a simple derivative shape:

$$B_0''''(r) = \frac{2[\alpha(1 + 3r^2) - \beta(r^4 - 3r^2 + 6)]}{(1 - r^2)^3}. \quad (64)$$

The numerator increases as a function of r^2 . Thus B_0'''' can change sign only from negative to positive. Since $B_0''''(r_0) < 0$, the same is true of B_0'''' . It follows that B_0'' has no interior maximum on any subinterval.

The fixed comparisons below establish

$$\begin{aligned} B_0'(r_0) &< -1/16, & B_0''(r_0) &< 1/3, & B_0'''(r_0) &< 0, \\ B_0''(4/5) &< 0, & B_0''(r_1) &< 0. \end{aligned}$$

The endpoint property gives $B_0'' < 1/3$ up to $4/5$, and $B_0'' < 0$ thereafter. Consequently

$$B_0'(r) < -\frac{1}{16} + \frac{4/5 - 5/8}{3} = -\frac{1}{240} < 0, \quad r_0 \leq r \leq 4/5,$$

and B_0' decreases after $4/5$. Therefore $B_0(r) \geq B_0(r_1) > 0$ throughout the interval. Equation (63) now gives CK on $[r_0, r_1]$.

G. Contract from the starting point to zero

The same estimate leaves extra slack at r_0 :

$$c_{\text{ent}}(r_0)(1 - r_0)(1 + r_0 - Mr_0^2) + r_0^2/2 - L > 0.$$

Thus $I_f(r_0) < r_0^2/2$. Entropy contraction gives

$$I_f(r) \leq (r/r_0)^2 I_f(r_0) \leq r^2/2 \leq \phi(r) \quad (0 \leq r \leq r_0).$$

The strict starting slack matters: contracting a bound equal to $\phi(r_0)$ would not suffice, since $\phi(r)/r^2$ increases. The local and large-bias alternatives were already covered. This completes the all-bias proof for $0 \leq r \leq \rho_*$.

H. The fixed comparisons

All coefficients above were defined by tangency. The following bounds only check finitely many values; they do not specify a fitted entropy curve. For $1 \leq y \leq 2$, put $z = (y - 1)/(y + 1)$ and define

$$\begin{aligned} \ell_-(y) &= 2 \sum_{j=0}^{11} \frac{z^{2j+1}}{2j+1}, \\ \ell_+(y) &= \ell_-(y) + \frac{2z^{25}}{25(1 - z^2)}. \end{aligned}$$

The logarithm series gives

$$\ell_-(y) \leq \log y \leq \ell_+(y).$$

For other positive rational arguments, first extract a power of two. Only the following logarithms occur:

$$\log(10/7), \quad \log(6/5), \quad \log(7/3), \quad \log 5.$$

Substituting the displayed sums gives

$$\alpha > \frac{53}{40}, \quad 0 < \beta < \frac{17}{20}, \quad z_0 < r_0^2, \quad r_1^2 < z_1.$$

These are the bounds used for the mean term, derivative shape, and interpolation domain. In the fixed checks below, evaluate α and β directly from their defining logarithms and the displayed finite sums; no rounded coefficient values are needed. For scale only, the same substitutions with outward rounding give

$$\begin{aligned} 1.3258315440444 &< \alpha < 1.3258315440447, \\ 0.8499134388946 &< \beta < 0.8499134388950. \end{aligned}$$

These decimal enclosures help a reader reproduce the calculation, but the proof uses the finite rational sums above.

For completeness, the derivatives needed in the five fixed checks are

$$\begin{aligned} B_0'(r) &= -2(1 + M)r + 3Mr^2 \\ &\quad + (\alpha - \beta r^2)g(r) + 2\beta rh(r), \\ B_0''(r) &= -2(1 + M) + 6Mr + \frac{\alpha - \beta r^2}{1 - r^2} \\ &\quad - 4\beta rg(r) + 2\beta h(r), \\ B_0'''(r) &= 6M - 6\beta g(r) - \frac{6\beta r}{1 - r^2} \\ &\quad + \frac{2r(\alpha - \beta r^2)}{(1 - r^2)^2}. \end{aligned}$$

At rational arguments, use $g(r) = \frac{1}{2} \log((1+r)/(1-r))$ and the definition of $h(r)$, replacing each logarithm by ℓ_- or ℓ_+ in the unfavorable direction. At the common boundary, the same exponent gives all three values:

$$r_1 = \frac{e^{31/10} - 1}{e^{31/10} + 1}, \quad g(r_1) = \frac{31}{20},$$

$$h(r_1) = \log(1 + e^{-31/10}) + \frac{31/10}{1 + e^{31/10}}.$$

The exponential sum and remainder in Section F-G bound the exponential. Substitution in these identities, followed by the logarithm bounds, gives the endpoint comparisons and $r_1^2 < z_1$. All substitutions are in the unfavorable direction in the displayed formula for B_0 and its derivatives. For the local margin, use the same bounds on r_1 and the logarithm formula for $h(5r_1/8)$. These finite sums establish the table, with $M = (7/8)^2$:

Fixed expression	Bound
$M_{\text{loc}}(r_1, 5/8)$	$> 1/1400$
$B'_0(5/8)$	$< -17/250$
$B''_0(5/8)$	$< 7/30$
$B'''_0(5/8)$	< -1
$B''_0(4/5)$	$< -1/9$
$B''_0(r_1)$	$< -1/10$
$B_0(r_1)$	$> 1/4000$
Starting contraction slack at $r_0 = 5/8$	$> 1/500$

These stronger rational margins are obtained by the same unfavorable finite-sum substitutions and exceed the weaker signs used in the proof. Every check uses a fixed logarithm or exponential value. The interval argument is the derivative shape in (64), not a grid of arithmetic checks.

APPENDIX E

ENDPOINT REDUCTION FOR THE HEIGHT PARAMETER

The scalar comparison initially depends on two variables: the channel's half-log-odds ℓ_ρ and one coordinate's comparison height v . This appendix removes the height variable analytically. The reason is elementary: a suitable upper bound can decrease and then increase, but cannot have an interior maximum. We only need its endpoints.

A. Two Shape Properties

Recall $c(v) = \arcsin(\tanh v)^2 / \tanh v$ and $F(v) = h(\tanh v) / \tanh v$. On $v > 0$,

$$c \text{ is concave,} \quad 1/F \text{ is convex.}$$

Here is a short proof. Put $s(v) = \arcsin(\tanh v) / \sinh v$. Differentiating gives

$$c'(v) = 2s(v) - s(v)^2, \quad 0 < s(v) < 1, \quad s'(v) < 0.$$

Indeed, $\sinh v - \arcsin(\tanh v)$ starts at zero and has positive derivative. Also

$$s'(v) = \frac{\tanh v - \arcsin(\tanh v) \cosh v}{\sinh^2 v} < 0,$$

since $\arcsin(\tanh v) > \tanh v > 0$. Hence $c''(v) = 2(1 - s(v))s'(v) < 0$, and $0 < c'(v) < 1$.

For the reciprocal profile write

$$J(v) = \frac{1}{F(v)} = \frac{\sinh v}{\mathcal{H}(v)}, \quad \mathcal{H}(v) = \cosh v h(\tanh v) > 0.$$

The difference $v \coth v - \log(2 \cosh v)$ decreases to zero: its derivative is $(\tanh v - v) / \sinh^2 v < 0$. Consequently $\mathcal{H}'(v) = \sinh v \log(2 \cosh v) - v \cosh v < 0$. Direct differentiation now gives

$$J'(v) = \frac{\log(2 \cosh v)}{\mathcal{H}(v)^2},$$

$$J''(v) = \frac{\tanh v \mathcal{H}(v) - 2 \log(2 \cosh v) \mathcal{H}'(v)}{\mathcal{H}(v)^3} > 0.$$

B. Lower and Upper Endpoints

Fix a channel with $\ell = \ell_\rho \geq 7/2$, write $r = \rho$, and use the canonical multiplier $p = (\lambda + 1/\alpha)^2$. The Parseval bound on the coordinate yield is

$$Y_P(v) = \frac{rc(v) + p[d_2/2 + (d_1 - d_2/2)F(\ell)/F(v)]}{v}.$$

We prove below that Y_P has no interior maximum when $pd_2/(2r) \geq 5/32$. Appendix F-C verifies this condition on the whole stated half-line. The LSI comparison covers heights through $z = \ell/2$. Define the last retained height V by $F(V) = F(\ell)/a_{\text{cut}}$, where the local lemma covers influences at least a_{cut} . If $V \leq z$, the LSI bound covers everything. Otherwise the only remaining checks are $Y_P(z) < B/\ell$ and $Y_P(V) < B/\ell$.

The log-sum argument handles the smaller channels without this coordinate comparison.

C. A Uniform Rate for the Entropy Profile

The cutoff proof needs no inverse formula for F . It only needs to know that F decreases fast enough. Define its relative decay rate $Q_F(v) = -F'(v)/F(v)$. We prove

$$Q_F(v) > 8/5 \quad (v > 0). \quad (65)$$

Writing $q_v = e^{-2v}$, differentiation of the entropy formula gives

$$Q_F(v) = \frac{4q_v[v + \log(1 + q_v)]}{(1 - q_v)[(1 + q_v) \log(1 + q_v) + 2vq_v]}.$$

After multiplication by positive denominators, the claimed bound is equivalent to

$$vq_v(1 + 4q_v) + (5q_v - 2 + 2q_v^2) \log(1 + q_v) > 0.$$

If the logarithm has nonnegative coefficient, this is immediate. Otherwise $\log(1 + q_v) < q_v$ reduces it to the stronger assertion $v - 2 + (4v + 5)e^{-2v} > 0$. This is immediate for $v \geq 2$. For $0 \leq v < 2$, the ratio $(2 - v)e^{2v}/(4v + 5)$ has derivative with the sign of $-8v^2 + 6v + 7$. Its maximum is therefore at $(3 + \sqrt{65})/8$, between $11/8$ and $7/5$. The maximum is less than $5e^{14/5}/84 < 1$. The last fixed exponential bound follows from the finite series and geometric remainder in Section F-G. This proves the rate.

For the cutoff $a_{\text{cut}} = 1/(1 + (8/3)e^{-\ell})$, integration between its actual height V and ℓ now gives

$$\begin{aligned} \frac{8}{5}(\ell - V) &< \int_V^\ell Q_F(v) dv = -\log a_{\text{cut}} \\ &= \log(1 + (8/3)e^{-\ell}) < \frac{8}{3}e^{-\ell}. \end{aligned}$$

Thus $0 < \ell - V < (5/3)e^{-\ell}$, without evaluating an inverse of F .

D. Endpoint Principle for the Original Yield

The original Parseval yield has no interior maximum once its constant term pays for the following global curvature bound:

$$\begin{aligned} \frac{-c''(v)[vJ'(v) - J(v)]}{J''(v)} - [c(v) - vc'(v)] &< \frac{5}{32}, \quad (66) \\ J &= 1/F, \quad v > 0. \end{aligned}$$

We prove it below. Its use is simple. For positive constants A_0, A_1 , the derivative of

$$Y_P(v) = \frac{rc(v) + A_0 + A_1J(v)}{v}$$

has the sign of

$$A_1 - \frac{A_0 + r[c(v) - vc'(v)]}{vJ'(v) - J(v)}.$$

The denominator is positive. Differentiation and (66) show that the fraction strictly decreases if $A_0/r \geq 5/32$. The derivative of Y_P can therefore change sign only from negative to positive: Y_P has no interior maximum. This applies directly to the Parseval yield with $A_0 = pd_2/2$ and $A_1 = p(d_1 - d_2/2)F(\ell)$. Appendix F-C proves $pd_2/(2r) > 3/16 > 5/32$ for every $\ell \geq 7/2$.

a) Small-height entropy comparison.: Recall $L = \log 2$. Continue to write $\mathcal{H}(v) = \cosh v h(\tanh v)$, and put $\Lambda(v) = \log(2 \cosh v)$. Recall that $\mathcal{H}' < 0$ and $J' = \Lambda/\mathcal{H}^2$. Put $r_v = \tanh v$, $\delta_v = 1 - r_v^2$. For a decreasing function f and an increasing function g on $[0, 1]$,

$$\begin{aligned} \int_0^1 fg - \left(\int_0^1 f \right) \left(\int_0^1 g \right) \\ = \frac{1}{2} \int_0^1 \int_0^1 [f(s) - f(t)][g(s) - g(t)] ds dt \leq 0. \end{aligned}$$

Apply this to $f(s) = 1/(1 + s)$, $g(s) = 1/(1 - r_v^2 s^2)$:

$$\frac{h(r_v)}{\delta_v} = \int_0^1 \frac{ds}{(1 + s)(1 - r_v^2 s^2)} \leq L \int_0^1 \frac{ds}{1 - r_v^2 s^2} = \frac{Lv}{r_v}.$$

The integral identity follows by integrating the partial fractions

$$\begin{aligned} \frac{1}{(1 + s)(1 - r^2 s^2)} &= \frac{1}{(1 - r^2)(1 + s)} \\ &\quad + \frac{r}{2(1 + r)(1 - rs)} \\ &\quad - \frac{r}{2(1 - r)(1 + rs)}. \end{aligned}$$

For $0 < v \leq 1$, this also gives $\mathcal{H}'' < 0$. Indeed, $v/\tanh v$ increases, so $h(r_v)/\delta_v \leq L \coth 1 < 1$: use $e^2 > 7$, hence $\coth 1 < 4/3$, and $L < 3/4$. Differentiation gives $\mathcal{H}'' =$

$(h(r_v) - \delta_v)/\sqrt{\delta_v}$. Thus $\mathcal{H}^{-2}$ is increasing and convex on this interval. The product rule gives $J''' = (\Lambda \mathcal{H}^{-2})'' > 0$, since Λ is positive, increasing, and convex. Hence

$$vJ'(v) - J(v) = \int_0^v tJ''(t) dt < \frac{v^2}{2}J''(v) \quad (0 < v \leq 1). \quad (67)$$

b) Accumulated curvature.: We first strengthen the preceding integral bound to

$$\frac{vJ'(v) - J(v)}{J''(v)} < \frac{v}{2} \quad (v > 0). \quad (68)$$

For $v \leq 1$, (67) already proves it. For $v \geq 1$, put

$$\begin{aligned} q &= e^{-2v}, & T &= \frac{\log(1 + q)}{q}, \\ K &= 2v + (1 + q)T, & D &= vJ' - J. \end{aligned}$$

The entropy formula gives $J = (1 - q)/(qK)$, and differentiation gives

$$\begin{aligned} qD &= 1 - \frac{1 - q + 2T}{K} + \frac{(1 - q^2)T^2}{K^2}, \\ K' &= 2T, \quad 0 < T' < q. \end{aligned}$$

Here primes denote derivatives in v . The last bound follows from the reciprocal trapezoid estimate: $T' = 2[T - 1/(1 + q)] < q$. Also $T \geq 1 - q/2$. The last numerator $(1 - q^2)T^2$ increases, so differentiating and dropping its positive derivative gives

$$\begin{aligned} (qD)' &\geq \frac{1}{K^2} \left[2T(1 - q + 2T) - K(2q + 2T') \right. \\ &\quad \left. - \frac{4(1 - q^2)T^3}{K} \right] > 0. \end{aligned}$$

The three terms are respectively greater than 4, less than 2, and less than 2. Indeed, $e^2 > 7$ gives $q < 1/7$ and $1 - q, T > 6/7$. Also $qK < e^{-2v}(2v + 3/2) < 1/2$, since this upper bound decreases from a value below $(1/7)(7/2)$ at one. Finally $K > 2$ and $T < 1$. Since $(qD)' = q(vJ'' - 2D)$, this proves (68).

c) Scaling of angular curvature.: Write

$$w(v) = \frac{\arcsin(\tanh v)}{\sinh v}, \quad k(v) = -c''(v) > 0.$$

The function w is decreasing and convex as a function of v^2 . For a direct proof, put $r_v = \tanh v$ and $\tau_v = \text{sech } v$. The elementary angle bound

$$\arcsin r_v \geq \frac{3r_v}{2 + \tau_v}, \quad w(v) \geq \frac{3\tau_v}{2 + \tau_v}$$

follows by differentiation: the derivative of the first difference, with respect to r_v , is $(1 - \tau_v)^2/[\tau_v(2 + \tau_v)^2] \geq 0$. Since $3\tau_v/(2 + \tau_v) > \tau_v^2$, the identity $w' = (\tau_v^2 - w)/r_v$ also proves that w decreases. Differentiate once more and substitute this lower bound:

$$vw'' - w' \geq \frac{\tau_v(1 - \tau_v)}{r_v^2(2 + \tau_v)} [(3 + \tau_v)r_v + v\{4 - (1 + \tau_v)^3\}] > 0.$$

For the last sign, $r_v \geq v\tau_v$ leaves $v(3 - 2\tau_v^2 - \tau_v^3) > 0$ inside the bracket. The sign $vw'' - w' > 0$ is exactly convexity in v^2 .

Consequently $f(v) = 1 - w(v)$ is increasing and concave in v^2 , and vanishes at zero. For $0 < t \leq v$, concavity gives

$f(t) \geq (t/v)^2 f(v)$ and $f'(t) \geq (t/v) f'(v)$. Since $k = 2ff'$, integration yields

$$c(v) - vc'(v) = \int_0^v tk(t) dt \geq \frac{v^2}{5} k(v). \quad (69)$$

Also $k(v) \leq \tanh^3(v)/2 < 1/2$: in $k = 2(1-w)(w-\delta)/r$, where $r = \tanh v$ and $\delta = 1 - r^2$, maximize the quadratic in w . Combining the two estimates gives

$$\begin{aligned} & \frac{k(v)[vJ'(v) - J(v)]}{J''(v)} - [c(v) - vc'(v)] \\ & \leq k(v) \left(\frac{v}{2} - \frac{v^2}{5} \right) < \frac{5}{32}. \end{aligned}$$

For $v \leq 5/2$, use $k < 1/2$ and complete the square; for larger v , the bracket is nonpositive. This proves (66), and hence the endpoint principle.

APPENDIX F

CANONICAL HIGH-CORRELATION COMPARISONS

The middle-range argument ends at $\ell = 7/2$. The following three comparisons cover the entire remaining half-line with the same canonical penalty $\gamma = \lambda\alpha$. The spectral multiplier is $p = (\lambda + 1/\alpha)^2$; it is not replaced by the angle constant $P = \pi^2/4$. The constant P still denotes the limit of the angle profile $c(v)$; it supplies elementary upper bounds and measures the angle saving at the half-height join. Keeping the exact spectral multiplier p preserves the low-degree cancellations in the join and cutoff proofs.

A. Reduction to Three Comparisons

Write $r = \rho = \tanh \ell$, $x = e^{-\ell}$, $L = \log 2$, and $\theta = \arcsin r$. Our definitions are

$$\begin{aligned} \lambda &= \theta/r, \quad qquad\alpha = (\theta + \sinh \ell)/\ell, \\ \gamma &= \lambda\alpha, \quad qquad d = \ell/2 + 5/4. \end{aligned} \quad (70)$$

As before, $\omega(s) = \gamma s/(\gamma + s)$, $d_1 = [\omega(d) - \omega(1)]r^2$, $d_2 = [\omega(d) - \omega(2)]r^4$, and $d_b = d_1 - d_2/2$. The retained-variance baseline is

$$\begin{aligned} p &= (\lambda + 1/\alpha)^2, & b &= \lambda^2(2 + 1/\gamma), \\ C_d &= p\omega(d) - b, & B &= 2\theta^2 + C_d\phi(r)/L. \end{aligned}$$

Here C_d is a channel-dependent spectral coefficient, not an entropy approximation constant. The degree balances the LSI exponential against the entropy ratio at half the channel height; the penalty itself is already fixed by the dictator-centered scalar tangent.

The local lemma handles influences at least $a_{\text{cut}} = 1/[1 + (8/3)x]$. For the remaining coordinates we prove:

- 1) The line-angle LSI yield is below B/ℓ for all heights up to $z = \ell/2$ (Section F-D).
- 2) The original Parseval yield is below B/ℓ at the same join (Section F-E).
- 3) That yield is below B/ℓ at the cutoff height $F(V) = F(\ell)/a_{\text{cut}}$ (Section F-F).

We first check the lower-variance sign, all-bias mean condition, and coefficient needed for the global endpoint principle. That

principle then fills the interval from z to V ; if $V \leq z$, the small-height comparison already covers it. No channel subdivision is needed in any of these checks.

B. Local Cutoff

Write $x = e^{-\ell}$. We use the single influence cutoff

$$a_{\text{cut}} = \frac{1}{1 + (8/3)x}. \quad (71)$$

We need this cutoff only for $\ell \geq 7/2$. On this half-line $x < 1/32$: indeed $e^3 > 20$ and $e^{1/2} > 1 + 1/2 + 1/8 = 13/8$ give $e^{7/2} > 32$. The proof retains the leading entropy logarithms until they cancel, then bounds the three remaining errors uniformly.

Put $r = (1 - x^2)/(1 + x^2)$, and let $h_b(p) = -p \log p - (1 - p) \log(1 - p)$ denote binary entropy in probability coordinates. The following temporary quantities expose the cancellation between the two entropies:

$$\begin{aligned} \kappa_{\text{loc}} &= \frac{4}{3 + 8x}, & \eta_{\text{loc}} &= \frac{3x}{4(1 + x^2)}, \\ p &= \frac{x^2}{1 + x^2}, & q_{\text{loc}} &= \kappa_{\text{loc}}x(1 + \eta_{\text{loc}}), \\ A_{\text{loc}} &= \frac{1}{2x^2} + \frac{3}{4x} + 1 + \frac{x^2}{2}. \end{aligned}$$

Thus $p = (1 - r)/2$, $q_{\text{loc}} = (1 - ra_{\text{cut}})/2$. The local margin $(1 - r^2)h(ra_{\text{cut}}) - (1 - a_{\text{cut}}r^2)h(r)$, divided by the positive factor $4\kappa_{\text{loc}}x^3/(1 + x^2)^2$, equals

$$\Lambda_{\text{loc}} = \frac{h_b(q_{\text{loc}})}{\kappa_{\text{loc}}x} - A_{\text{loc}}h_b(p).$$

a) *Cancellation of the leading logarithms.*: Both entropy estimates follow from one pair of logarithm bounds:

$$\frac{2t}{2+t} \leq \log(1+t) \leq t - \frac{t^2}{2(1+t)} \quad (t \geq 0).$$

These are the midpoint and trapezoid bounds for the integral of the convex function $s \mapsto (1 + ts)^{-1}$ on $[0, 1]$, multiplied by t . The lower bound, with $t = q_{\text{loc}}/(1 - q_{\text{loc}})$, gives

$$\begin{aligned} h_b(q_{\text{loc}}) &\geq q_{\text{loc}} \log(1/q_{\text{loc}}) + q_{\text{loc}} \\ &\quad - \frac{q_{\text{loc}}^2}{2 - q_{\text{loc}}}. \end{aligned}$$

The upper bound gives $(1 + \eta_{\text{loc}}) \log(1 + \eta_{\text{loc}}) \leq \eta_{\text{loc}} + \eta_{\text{loc}}^2/2$. Consequently

$$\begin{aligned} \frac{h_b(q_{\text{loc}})}{\kappa_{\text{loc}}x} &\geq (1 + \eta_{\text{loc}})(\ell - \log \kappa_{\text{loc}}) + 1 \\ &\quad - \frac{\eta_{\text{loc}}^2}{2} - \frac{\kappa_{\text{loc}}x(1 + \eta_{\text{loc}})^2}{2 - q_{\text{loc}}}. \end{aligned}$$

Here $\eta_{\text{loc}} \leq 3x/4$ and $q_{\text{loc}} \leq x(4 + 3x)/(3 + 8x)$. Keep their two errors together:

$$\begin{aligned} & \frac{\eta_{\text{loc}}^2}{2} + \frac{\kappa_{\text{loc}}x(1 + \eta_{\text{loc}})^2}{2 - q_{\text{loc}}} \\ & \leq \frac{9}{32}x^2 + \frac{x(4 + 3x)^2}{12(2 + 4x - x^2)} \leq \frac{2}{3}x + \frac{1}{5}x^2. \end{aligned}$$

For the last inequality, the identity

$$(8-x)(2+4x-x^2) - (4+3x)^2 = x(6-21x+x^2) > 0$$

holds for $x < 1/4$. It bounds the rational term by $2x/3 - x^2/12$; then $9/32 - 1/12 < 1/5$.

For the other entropy, write $h_b(p) = \log(1+x^2) + 2\ell x^2/(1+x^2)$. The same trapezoid bound for the logarithm yields

$$\begin{aligned} A_{\text{loc}} \log(1+x^2) &\leq \frac{1}{2} + \frac{3}{4}x + \frac{3}{4}x^2 \\ &\quad + \frac{x^3[2x(1+x^2) - 3]}{8(1+x^2)} \\ &\leq \frac{1}{2} + \frac{3}{4}x + \frac{3}{4}x^2. \end{aligned}$$

The remainder is negative for $x < 1/4$. Therefore

$$\begin{aligned} A_{\text{loc}} h_b(p) &\leq \frac{1}{2} + \frac{3}{4}x + \frac{3}{4}x^2 \\ &\quad + \frac{2A_{\text{loc}}\ell x^2}{1+x^2}. \end{aligned}$$

The coefficient of ℓ now cancels exactly to

$$1 + \eta_{\text{loc}} - \frac{2A_{\text{loc}}x^2}{1+x^2} = -x \left[\frac{3}{4(1+x^2)} + x \right].$$

Therefore

$$\begin{aligned} \Lambda_{\text{loc}} &\geq \frac{1}{2} - (1 + \eta_{\text{loc}}) \log \kappa_{\text{loc}} - \frac{17}{12}x - \frac{19}{20}x^2 \\ &\quad - \ell x \left[\frac{3}{4(1+x^2)} + x \right]. \end{aligned} \quad (72)$$

b) Uniform remainder estimates.: The same trapezoid bound at $t = 1/3$ gives $\log(4/3) < 7/24$. Since $\eta_{\text{loc}} \leq 3/128$ and $\kappa_{\text{loc}} < 4/3$,

$$(1 + \eta_{\text{loc}}) \log \kappa_{\text{loc}} < \left(1 + \frac{3}{128}\right) \frac{7}{24} < \frac{1}{3}.$$

Also

$$\frac{17}{12}x + \frac{19}{20}x^2 < \frac{17}{12 \cdot 32} + \frac{19}{20 \cdot 32^2} < \frac{1}{20}.$$

Finally $\ell e^{-\ell}$ decreases here, so $\ell x < 7/64$, and

$$\ell x \left[\frac{3}{4(1+x^2)} + x \right] < \frac{7}{64} \left(\frac{3}{4} + \frac{1}{32} \right) < \frac{3}{32}.$$

Substitution in (72) proves

$$\begin{aligned} \Lambda_{\text{loc}} &> \frac{1}{2} - \frac{1}{3} - \frac{1}{20} - \frac{3}{32} \\ &= \frac{11}{480} > 0. \end{aligned}$$

Thus the criterion holds at a_{cut} throughout $\ell \geq 7/2$. For fixed r , its left side $(1-r^2)h(ra) - (1-ar^2)h(r)$ is concave in a and vanishes at $a = 1$. It is therefore nonnegative for every $a \in [a_{\text{cut}}, 1]$.

C. Canonical Penalty and Preliminary Bounds

Throughout this appendix $\ell \geq 7/2$, $x = e^{-\ell}$, and $\theta = \arcsin r$. The rule is

$$\begin{aligned} \lambda &= \theta/r, & \alpha &= (\theta + \sinh \ell)/\ell, \\ \gamma &= \lambda\alpha, & d &= \ell/2 + 5/4, \end{aligned}$$

and

$$\begin{aligned} p &= (\lambda + 1/\alpha)^2, & b &= \lambda^2(2 + 1/\gamma), \\ C_d &= p\omega(d) - b, & B &= 2\theta^2 + C_d\phi(r)/L. \end{aligned}$$

The elementary bounds $e^{7/2} > 33$, $157/50 < \pi < 22/7$, and $2/3 < L < 3/4$ give $r > 99/100$, $\theta > 3/2$, and $\lambda^2 < P < 5/2$; the angle ratio is bounded by the endpoint chord of the convex function $\arcsin r$. Since $\theta + \sinh \ell > 1/(2x)$,

$$\gamma > \frac{3e^\ell}{4\ell} > \ell + \frac{7}{2} = 2d + 1. \quad (73)$$

For the second inequality, $3e^\ell - 4\ell^2 - 14\ell$ is positive at $7/2$, its first derivative is positive there, and its second derivative $3e^\ell - 8$ stays positive.

The exact multiplier identities are

$$\begin{aligned} p\omega(1) - b &= -\lambda^2, \\ p\omega(2) - b &= -\frac{\lambda^2}{\gamma + 2}, \\ C_d &= \lambda^2 \left[d - 2 - \frac{(d-1)^2}{\gamma + d} \right]. \end{aligned} \quad (74)$$

Thus $C_d > 0$, since $\gamma(d-2) > 1$, and $C_d < (5/2)(d-2)$, $b < (5/2)(2 + 1/7) < 6$. The all-bias mean condition follows without a separate channel test:

$$\begin{aligned} \gamma - b - C_d \left(1 - \frac{1}{2L} \right) &> \ell + \frac{7}{2} - 6 - \frac{5}{6} \left(\frac{\ell}{2} - \frac{3}{4} \right) \\ &= \frac{1}{6} + \frac{7}{12} \left(\ell - \frac{7}{2} \right) > 0. \end{aligned}$$

Here $0 < 1 - 1/(2L) < 1/3$. Finally, the coefficient required by the original-yield endpoint principle satisfies

$$\begin{aligned} \frac{pd_2}{2r} &= \frac{p}{2}(d-2) \frac{\gamma}{\gamma+d} \frac{\gamma}{\gamma+2} r^3 \\ &> 1 \cdot 1 \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{3}{4} = \frac{3}{16} > \frac{5}{32}. \end{aligned}$$

Also $0 < d_2 < d_1$, so $d_b > 0$. The global shape inequality (66) therefore applies with multiplier p .

a) Entropy estimate.: Put $E = h(r) - L(1-r^2) \geq 0$. For every $\ell \geq 3/2$,

$$E \leq x^2(2\ell - 5/3). \quad (75)$$

Indeed, with $q = x^2$, the entropy formula gives

$$2\ell - \frac{E}{q} = \frac{2\ell q}{1+q} + \frac{4L}{(1+q)^2} - \frac{\log(1+q)}{q}.$$

The reciprocal trapezoid bound gives $\log(1+q)/q \leq (2+q)/[2(1+q)]$. Using $2\ell \geq 3$ leaves

$$2\ell - \frac{E}{q} - \frac{5}{3} \geq \frac{4(L-2/3) - 11q/6 + 5q^2/6}{(1+q)^2} > 0.$$

The last numerator exceeds $8/81 - 11/120 > 0$: the first two terms of $L = 2 \operatorname{atanh}(1/3)$ give $L - 2/3 > 2/81$, and $e^3 > 20$ gives $q < 1/20$.

D. Small Heights: Paired Spectral Costs

Let $z = \ell/2$. The line-angle LSI comparison is

$$M_{\text{small}} = \frac{B}{\ell} - r - \frac{pK_{\text{LS}}r^2F(\ell)}{zF(z)},$$

$$K_{\text{LS}} = \frac{\gamma^2 e^{2d-3}}{2(\gamma+d)(\gamma+1)}.$$

Set $a_d = d - 1$, $k = e^{-1/2}$, $T = e^\ell F(\ell)/F(z)$, and $f_d = C_d/\lambda^2$. The identities (74) and $\phi(r)/L = r^2 - E/L$ give the useful cancellation

$$\frac{\ell M_{\text{small}}}{\theta^2} = d - \frac{\ell}{c(\ell)} - kT - \frac{a_d(a_d - kT)}{\gamma + d} - \frac{f_d E}{Lr^2}. \quad (76)$$

We bound the first three terms below, keeping the resolvent loss paired with the profile cost in the fourth term.

The elementary logarithm bounds $t/(1+t) \leq \log(1+t)$ and $(1+t)\log(1+t) \leq t + t^2/2$, obtained by integration, yield

$$\frac{2\ell + 1}{(\ell + 1 + x/2)(1+x)} \leq T \leq \frac{2\ell + 1 + x^2/2}{(\ell + 1)(1+x)}$$

$$\leq 2 - \frac{1}{\ell + 1}.$$

The lower bound exceeds $5/3$: after multiplication by its positive denominator, the difference has numerator

$$\ell - 2 - 5\ell x - \frac{15}{2}x - \frac{5}{2}x^2 \geq \frac{3}{2} - \frac{25}{32} - \frac{5}{2048} > 0.$$

Here this expression increases with ℓ and decreases with x on $\ell \geq 7/2$, $0 < x < 1/32$. Since $3/5 < k < 5/8$ and $c(\ell) > \theta^2 > 9/4$,

$$1 < kT < \frac{5}{8} \left(2 - \frac{1}{\ell + 1} \right),$$

$$d - \frac{\ell}{c(\ell)} - kT > \frac{\ell}{18} + \frac{5}{8(\ell + 1)} \geq \frac{1}{3}.$$

The last difference from $1/3$ equals $(2\ell - 3)(2\ell - 7)/[72(\ell + 1)]$.

Next, $\gamma + d > 4(d - 1)(d - 2)$. By (73), it suffices that $3e^\ell > 4\ell^3 - 6\ell^2 - 8\ell$. For $s = \ell - 7/2 \geq 0$, the positive cubic Taylor sum for e^s and $e^{7/2} > 33$ bound this difference below by

$$29 + 2s + \frac{27}{2}s^2 + \frac{25}{2}s^3 > 0.$$

Consequently

$$\frac{a_d(a_d - kT)}{\gamma + d} < \frac{a_d(a_d - 1)}{\gamma + d} < \frac{1}{4}.$$

Finally, (75), $f_d < d - 2$, $L > 2/3$, and $r^2 > 3/4$ give

$$\frac{f_d E}{Lr^2} < 2(d - 2)(2\ell - 5/3)e^{-2\ell} < \frac{1}{96}.$$

The product decreases: its logarithmic derivative is at most $1/2 + 3/8 - 2 < 0$. At $7/2$ its value is less than $2(16/3)/32^2 = 1/96$. Thus

$$M_{\text{small}} > \frac{\theta^2}{\ell} \left(\frac{1}{3} - \frac{1}{4} - \frac{1}{96} \right) = \frac{7\theta^2}{96\ell} > 0.$$

Since $vF(v)$ decreases and $c(v) \leq v$, this comparison at z covers every $0 < v \leq z$. In particular $B/\ell > r$.

E. Half-Height Join and Angular Saving

Write $a_z = F(\ell)/F(z)$. The original Parseval margin at the join has the exact form

$$M_{\text{join}} := Bz/\ell - rc(z) - p[d_2/2 + d_b a_z]$$

$$= r[c(\ell) - c(z)] - \frac{\lambda^2 r^4}{2(\gamma + 2)} + \frac{C_d}{2} \left[1 - r^4 - \frac{h(r)}{L} \right] - pd_b a_z.$$

The multiplier identities also give

$$pd_b = C_d(r^2 - r^4/2) + \lambda^2 \left[r^2 - \frac{r^4}{2(\gamma + 2)} \right] \quad (77)$$

$$\leq C_d/2 + P \leq Pd/2.$$

The angle saving is of order $\sqrt{x}$, while the losses are of order x . To see it directly, the function $e^v[P - c(v)]$ increases. With $t = \arcsin(\tanh v)$,

$$(t + \cos t)^2 - c(v) - c'(v)$$

$$= (1 - \sin t)(t/\sin t - \cos t)^2 + \sin t \cos^2 t > 0.$$

As $t + \cos t < \pi/2$, this gives $c + c' < P$. At $v = \log(10/3)$, the bounds $4/\pi > 5/4$ and $\arctan y \geq y - y^3/3$ at $y = 3/10$ give

$$2t/\pi < 16/25, \quad c(v)/P < (16/25)^2(109/91) < 1/2.$$

Hence $P - c(z) > (5P/3)\sqrt{x}$, since $z \geq 7/4 > \log(10/3)$. On the other hand, $c(\ell) \geq \theta^2 \geq P - 2\pi x$, so

$$c(\ell) - c(z) > \frac{5P}{3}\sqrt{x} - 2\pi x.$$

Also $h(r) < (2\ell + 1)x^2$, and $\log(1+x) \geq x/(1+x)$ gives

$$\frac{F(\ell)}{F(z)} < x \frac{2\ell + 1}{\ell + 1} \frac{1 + x^2}{1 + x} < 2x.$$

Using these bounds, (77), and (73), and discarding the positive $C_d(1 - r^4)/2$, gives

$$\frac{M_{\text{join}}}{P\sqrt{x}} > \frac{5r}{3} - \left(\frac{7}{6}\ell + \frac{5}{4} + \frac{8}{\pi} \right) e^{-\ell/2}$$

$$- \frac{(d-2)(2\ell+1)}{2L} e^{-3\ell/2}.$$

The first term increases and both losses decrease. For the affine loss this follows by differentiation; for the last, the logarithmic derivative is $1/(\ell - 3/2) + 1/(\ell + 1/2) - 3/2 < 0$. At $\ell = 7/2$, use $r > 99/100$, $\sqrt{x} < 7/40$, $8/\pi < 8/3$, and $2L > 4/3$. The affine coefficient is less than eight, and the last loss is less than $6(7/40)^3 < 1/25$. Thus the whole half-line satisfies

$$\frac{M_{\text{join}}}{P\sqrt{x}} > \frac{33}{20} - \frac{7}{5} - \frac{1}{25} = \frac{21}{100} > 0.$$

F. Cutoff Bound with Exact Low-Degree Multipliers

The local cutoff is $a_{\text{cut}} = 1/[1 + (8/3)x]$. Let $F(V) = F(\ell)/a_{\text{cut}}$. The global profile bound $-F'/F > 8/5$ and $\log(1+t) < t$ give

$$0 < \ell - V < \frac{5}{3}x.$$

We check the actual Parseval margin

$$M_{\text{cut}} = \frac{B}{\ell}V - rc(V) - p[d_2/2 + d_b a_{\text{cut}}].$$

By (74), its exact dictator deficit is

$$B - pd_1 - rc(\ell) = C_d[\phi(r)/L - r^2] = -C_d E/L.$$

Concavity of c , together with $c' > 0$, bounds the displacement loss below by $-(B/\ell)(\ell - V)$. Since $B > 0$,

$$M_{\text{cut}} \geq -C_d E/L + pd_b(1 - a_{\text{cut}}) - \frac{B}{\ell}(\ell - V).$$

This avoids any separate sign condition on a tangent coefficient. Use $B \leq 2\lambda^2 r^2 + C_d$, (77) as an equality in its first line, and (75). They give

$$\begin{aligned} \frac{M_{\text{cut}}}{x} &> C_d \left[\frac{8r^2(1 - r^2/2)}{3 + 8x} - \frac{(2\ell - 5/3)x}{L} - \frac{5}{3\ell} \right] \\ &\quad + \lambda^2 r^2 \left[\frac{8}{3 + 8x} \left(1 - \frac{r^2}{2(\gamma + 2)} \right) - \frac{10}{3\ell} \right]. \end{aligned}$$

Both brackets have uniform positive lower bounds. Indeed $x < 1/32$ implies $r^2 > 3/4$, so

$$\frac{8r^2(1 - r^2/2)}{3 + 8x} > \frac{32}{13} \frac{15}{32} > 1.$$

The function $(2\ell - 5/3)e^{-\ell}$ decreases; using $L > 2/3$,

$$\frac{(2\ell - 5/3)x}{L} < \frac{16}{3} \frac{1}{32} \frac{3}{2} = \frac{1}{4}, \quad \frac{5}{3\ell} \leq \frac{10}{21} < \frac{1}{2}.$$

Thus the first bracket exceeds $1/4$. For the second, $8/(3+8x) > 2$, $1 - r^2/[2(\gamma + 2)] > 3/4$, and $10/(3\ell) \leq 20/21 < 1$. It therefore exceeds $1/2$. We conclude on the entire half-line that

$$M_{\text{cut}} > x \left(\frac{C_d}{4} + \frac{\lambda^2 r^2}{2} \right) > 0.$$

This completes the third comparison. The original-yield endpoint principle now fills the retained heights between the join and the cutoff; heights below the join were covered by the LSI comparison.

G. Fixed Constants

All fixed exponential comparisons follow from one positive sum and its geometric remainder: for $0 < y < 14$,

$$E_{12}(y) := \sum_{j=0}^{12} \frac{y^j}{j!} < e^y < E_{12}(y) + \frac{y^{13}}{13!(1 - y/14)}.$$

This proves the exponential brackets used above by rational substitution.

For logarithms use, for $|y| < 1$,

$$\begin{aligned} \log \frac{1+y}{1-y} &= 2 \sum_{j=0}^{N-1} \frac{y^{2j+1}}{2j+1} + R_N(y), \\ |R_N(y)| &\leq \frac{2|y|^{2N+1}}{(2N+1)(1-y^2)}. \end{aligned}$$

Taking $y = 1/3$ bounds $\log 2$; taking $y = (q-1)/(q+1)$ bounds $\log q$ for every positive rational q . The low-correlation arithmetic uses $N = 12$. The basic constant bounds are

$$3.14159 < \pi < 3.1416, \quad .69314 < \log 2 < .69315.$$

For π , use Machin's identity

$$\pi = 16 \arctan(1/5) - 4 \arctan(1/239).$$

Bound the first arctangent between its first four and first five alternating terms, and the second between its first two and first three. To verify the identity, the tangent addition formula gives

$$\tan(4 \arctan(1/5)) = 120/119;$$

subtracting $\arctan(1/239)$ gives tangent one and an angle in $(0, \pi/2)$. For $\log 2$, take the first five terms of the displayed logarithm series at $y = 1/3$ and bound the remainder by

$$\frac{2(1/3)^{11}}{11(1 - 1/9)}.$$

In particular these imply the bounds used here, $25/8 < \pi < 22/7$, $9/4 < P < 5/2$, and $2/3 < L < 3/4$. The positive exponential sums give $e^3 > 20$, $e^{7/2} > 33$, and $3/5 < e^{-1/2} < 5/8$. Square-root bounds follow by squaring positive rational endpoints.

a) Completion. The three comparisons, the mean condition, and the local cutoff cover every $\ell \geq 7/2$, including the shared endpoint. All estimates are printed above; no interval receipt, symbolic sign certificate, or external computation supplies a missing comparison.

APPENDIX G

NUMERICAL VERIFICATION AND REPRODUCIBILITY

This appendix records independent numerical checks of the quantities used in the proof. They are designed to catch transcription, normalization, and sign errors. They are not proof inputs: finite enumeration cannot cover all dimensions, and a mesh cannot establish an inequality on a continuum. The analytic arguments in the preceding appendices remain the certificates.

Remark 24 (Reproducing the diagnostics). From the source directory, run

```
python3 numerical_diagnostics.py.
```

The script uses NumPy for finite-cube enumeration and stable elementary formulas for the scalar channel quantities. Information values are in nats; scalar margins use the normalizations of the cited lemmas. Displayed values are rounded only after the underlying expressions have been evaluated.

A. Exhaustive Checks on Small Cubes

For a truth table f on n bits, the computation forms the full noise matrix

$$K_\rho(y, x) = 2^{-n} \prod_{i=1}^n (1 + \rho x_i y_i),$$

$$u(y) = \sum_x K_\rho(y, x) f(x),$$

and evaluates $I_f(\rho) = h(\mathbb{E}f) - 2^{-n} \sum_y h(u(y))$. Thus there is no Monte Carlo error. We enumerated all $2^{16} = 65,536$ Boolean truth tables for $n = 4$. We also enumerated all 7,581 monotone truth tables for $n = 5$, generating them recursively from lower and upper sections $g \leq h$.

At every tested channel the largest lead $I_f - \phi$ was attained by a coordinate rule and differed from zero by at most 3.4×10^{-16} . The table gives the positive deficit $\phi - I_f$ of the best rule after excluding dictators and anti-dictators in dimension four, and after excluding coordinate rules in the monotone dimension-five enumeration.

ρ	all rules, $n = 4$	monotone rules, $n = 5$
0.10	8.70084×10^{-4}	5.09022×10^{-4}
0.25	5.33512×10^{-3}	3.15088×10^{-3}
0.50	1.94985×10^{-2}	1.18769×10^{-2}
0.75	3.37645×10^{-2}	2.12692×10^{-2}
0.90	3.11339×10^{-2}	1.93117×10^{-2}
0.99	1.04605×10^{-2}	6.06839×10^{-3}

These deficits are not uniform as ρ approaches either endpoint; that is expected and is one reason the proof uses an endpoint argument rather than a global numerical gap.

B. Low-Correlation Certificates

The two tangent coefficients recompute as

$$\alpha = 1.325831544044512 \dots,$$

$$\beta = 0.849913438894783 \dots$$

Substitution into the expressions of Section D-H gives the following values. The last column is the rational certificate printed in the proof.

Expression	Recomputed value	Certificate
$M_{\text{loc}}(r_1, 5/8)$	8.014515×10^{-4}	$> 1/1400$
$B'_0(5/8)$	-6.848011×10^{-2}	$< -17/250$
$B''_0(5/8)$	2.332248×10^{-1}	$< 7/30$
$B'''_0(5/8)$	-1.029815	< -1
$B''_0(4/5)$	-1.196782×10^{-1}	$< -1/9$
$B''_0(r_1)$	-1.123197×10^{-1}	$< -1/10$
$B_0(r_1)$	3.913927×10^{-4}	$> 1/4000$
Starting contraction slack	2.473064×10^{-3}	$> 1/500$

Every recomputed value lies on the strict side of its rational bound. The narrowest displayed clearance is the curvature check at $5/8$: $7/30 - B''_0(5/8) = 1.08491 \times 10^{-4}$.

C. Intermediate-Range Reserves

In this subsection, $\alpha, \beta, \gamma, \lambda, \eta$ are the middle-range parameters of Lemma 22; α and β no longer denote the tangent

coefficients above. For reference, define the half-height profile margin and the two channel reserves by

$$R_{\text{prof}}(\ell) = \log \frac{(\ell/2)F(\ell/2)}{\ell F(\ell)} - \beta[H(\ell) - H(\ell/2)],$$

$$R_{\text{en}}(\ell) = \theta^2 - rH(\ell) - \eta/2,$$

$$R_{\text{mean}}(\ell) = \gamma - \frac{2\theta^2}{\sinh^2 \ell} - \frac{\lambda}{\alpha} - r\ell.$$

These are direct evaluations of the three comparisons in Lemma 22; the full profile comparison has equality at $v = \ell$, so R_{prof} is shown at $v = \ell/2$.

ℓ	R_{prof}	R_{en}	R_{mean}
1.55	0.436092	0.752286	0.252406
2.00	0.607533	0.809600	0.593983
2.50	0.800374	0.691309	1.201045
3.00	1.001815	0.424392	2.248303
3.50	1.215187	0.057980	4.001328

A supplementary mesh used 196 channel heights at spacing 0.01 on $[1.55, 3.5]$ and the 99 ratios $v/\ell \in \{0.01, \dots, 0.99\}$. No negative profile margin was found; the smallest sampled one was 8.2344×10^{-3} , adjacent to the equality endpoint. The smallest sampled energy and mean reserves were respectively 0.057980 and 0.252406, matching the opposite ends of the interval in the table.

D. High-Correlation Margins

The raw join and cutoff margins tend to zero as ℓ grows. To distinguish genuine loss of sign from the expected asymptotic scale, the table reports $e^{\ell/2} M_{\text{join}}$ and $e^\ell M_{\text{cut}}$, together with the unscaled small-height margin.

ℓ	M_{small}	$e^{\ell/2} M_{\text{join}}$	$e^\ell M_{\text{cut}}$
3.5	0.158997	2.403324	4.548284
5	0.202903	3.896735	8.236764
10	0.253236	5.976483	18.337516
20	0.241820	6.279837	35.114182

On a 0.01-spaced mesh of 1,651 heights from $7/2$ to 20, all three quantities stayed positive. Their smallest sampled values were 0.158997, 2.403324, and 4.548284, all at the shared endpoint $\ell = 7/2$. This agrees with the analytic proof's choice to preserve the different asymptotic scales rather than seek a uniform positive lower bound for the raw margins.

a) Outcome. The exhaustive enumerations found no finite-dimensional counterexample, and the independently evaluated low, middle, and tail quantities showed no sign or normalization discrepancy. This is useful corroboration, but the theorem rests only on the analytic certificates printed earlier.

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